Supply Chain of Construction Materials in Earthquake Affect Districts- Nuwakot and Rasuwa



Assessment Report

Practical Action
South Asian Regional Office, Nepal

Supply chain of construction materials in earthquake affected districts- Nuwakot and Rasuwa

Assessment Report

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1. Introduction

1.1 Background:

The 2015 Nepal earthquake caused widespread destruction. Nearly 500,000 houses were destroyed and more than 250,000 houses were partially damaged. More than 70 per cent of the affected people are from remote hilly districts with poor socio-economic background. As per the Post Disaster Need Assessment (PDNA) report 609,938 houses need to be re-constructed and are mostly located in remote hilly areas.
The National Reconstruction Authority (NRA) is promoting Department of Urban Development and Building Construction (DUDBC) house designs for earthquake-resilient houses and construction guidelines. For this, ensuring access of quality construction materials and tools as per the required quantity at affordable prices in rural areas is a key challenge. In this context, this study is focused on assessing the demand and supply situation of construction materials and tools in Nuwakot and Rasuwa districts and explores key bottlenecks in the existing market system of construction materials.

1.2 Objectives and Scope of the Study

The overall objective of the assessment is to assess existing situation and key issues in supply chain of construction material in the project districts- Nuwakot and Rasuwa.
The followings are the specific objectives:

- Assess demand and supply situation of different construction materials
- Identify supply chain actors, their function and business relationship
- Assess capacity of the existing vendors and producers to meet the existing and future demand of the construction materials
- Understand supply chain economics (margins, profits, costs and credit terms)
- Assess available natural resources (raw materials for the construction materials) in the study districts and explore opportunities and constraints to promote local micro-enterprise on construction materials
- Identify major bottlenecks in the supply chain of the construction materials
- Identify major policy and legal barriers in improving supply system of construction materials

1.3 Description of Study Areas

Nuwakot: Nuwakot district is located in Bagmati zone in Mid Development Region of Nepal. It covers an area of 1121 square kilometre within latitude 27°48' N to 28°06' N and longitude 84°58'E to 85°30''E with an altitude from 518m to 4876m. The district is bordered with Dhading, Sindhupalchowk, Kathmandu and Rasuwa districts. There are 61 VDCs and 1 municipality in Nuwakot district. According to the national census 2011, the total population of the district is 277471 comprising 132787 male and 144684 female. Besides the agriculture farming, small scale livestock is the main source of occupation and livelihood of the majority of population.
Nuwakot district is served by surface transport facilities linking the district with the national strategic road network through Prithivi Highway, Narayanghad- Muglin- Galchi –Kathmandu. The network of feeder roads, district roads and village roads are increasing significantly in the district. However, the district and village roads are mostly in fair or poor conditions which require upgrading/rehabilitation and proper maintenance.
As a result of the devastating earthquake in 2015 and the aftershocks that followed, 79762 private houses were reported as damaged. Out of this 75562 houses were fully damaged and 4200 houses were partially damaged. Additionally, it was also reported that 485 school buildings, 99 health facility buildings and 29 government buildings were damaged due to the earthquake. In regards to human casualties, it was reported that 1109 people lost their lives and another 1050 were injured in the district alone.
Birgunj, Bhairawa, Butwal and Narayangrah are the main regional hubs from where construction materials are supplied to Nuwakot district. Bidur Municipality, headquarter of the district, is the main supply point within the

district. Three main markets namely Batar, Bidur and Trisuli exist within Bidur Municipality to supply construction materials to villages.

Rasuwa: Rasuwa is one of the eight districts of the Bagmati zone. It is bordered by China in the north, Dhadhing in the west, Nuwakot in the south, and Nuwakot and Sindhupalchowk in the east. The district headquarters is Dhunche. The district covers an area of 1544 sq.km. The elevation ranges from 905 to 7,408 metres, with several peaks above 7,000 metres. The main caste/ethnic groups are Tamang (64%) and Brahmin (16%). According to the national census 2011, the total population of the district is 43300 comprising 21475 male and 21825 female. There are 18 VDCs but no municipality in Rasuwa district. Located in the central part of Nepal, Rasuwa district is one of the richest districts in terms of natural, religious as well as cultural heritage. Some of the most popular tourist destinations like *Langtang* and *Gosaikund* are also found in the district.

As a result of the devastating earthquake in 2015 and the aftershocks that followed, 11635 private houses were reported as damaged. Out of this 11368 houses were fully damaged and 267 houses were partially damaged. Additionally, it was also reported that 98 school buildings, 27 health facility buildings and 12 government buildings were damaged due to the earthquake. In regards to human casualties, it was reported that 660 people lost their lives and another 771 were injured in the district alone.

Trisuli, Battar and Bidur are the main supply points of construction materials for Rasuwa district. Narayangarh and in few cases Kathmandu are the main regional hubs.

2. Methodology

An exploratory survey research method was used for the assessment. Various tools were used to collect quantitative and qualitative information. As no single source would provide complete information of the supply chain of the construction materials, the data were collected from multiple stakeholders. The method and various data collection tools used in the assessment are discussed below.

2.1 Desk Review

A review of existing literature on Sustainable Supply Chain Management (SSCM) of construction materials was conducted to inform the study design. Though there found limited literatures on the supply chain of construction materials for rural housing, the study has been hugely benefited from the review of supply chain analysis of rural sanitation products (latrine) conducted in different part of the world. Likewise, review of Post Disaster Need Assessment (PDNA) and Post Disaster Recovery Framework (PDRF) Nepal was helpful to understand the focus of the study.

2.2 Preliminary field visit

The study team made a four-day visit of Nuwakot and Rasuwa districts in the month of June 2016 to have a bird's eye view of the reconstruction and the supply chain situation in the districts. The team interacted with Chief District Officer (CDO), Local Development Officer (LDO) and representative of district Chamber of Commerce and Industries (CCI) of both the districts and the chief of sub regional NRA office in Nuwakot. The interaction was followed by field visit to some of the market centers of the districts and interaction with vendors. The visit provided general overview of the reconstruction situation in the district and market centers in the district. It was useful for division of districts into study clusters and sub-clusters and developing the data collection strategy.

2.3 Clusters /Sub-clusters division

During the field visit the study team observed that the big vendors in Nuwakot district are concentrated at Bidur, the district head quarter. Bidur, in fact, is catering the majority of the construction materials need of Rasuwa district as well (as no significant construction materials vendors even in Dhunche are present). Only small vendors with limited capacity are present in other part of the districts resulting high dependency on Bidur for construction materials. It has not only made accessing the construction materials difficult but also increased the transaction cost for people living in the other part of the districts. The study team realized that people in other parts of the districts will also have easy access to the material only if the materials are available at the local markets. For developing local market centers as the local construction material hubs, it is imperative to understand the demand and supply at the cluster. Hence, the study team decided to take the cluster as the unit of the study. The cluster is defined as the areas served by the market center. Accordingly, Nuwakot district was divided into 9 clusters and Rasuwa was divided into 6 clusters. Hence, for the cluster division, first of all the prominent market centers in each district were identified and the VDCs served by the market centers were demarcated.

The VDCs within the clusters have different level of access to road and market, which would ultimately affect on the access and choice of the construction materials. In this context, each cluster is further divided into three sub-clusters depending upon how close or far the VDCs of each cluster are located from the main road. For convenience, the sub-clusters were named as u*rban type (near the main road), semi-urban type (a little far away from the main road), and rural type (very far away from the main road).*The following figures present the name of clusters and the number of VDC/NP within each sub cluster. Nuwakot district was divided into 9 Clusters which were further divided into 62 sub-clusters. The sub clusters constitutes 6 Urban, 36 Semi urban and 20 rural. Similarly, Rasuwa district was divided into 6 clusters, which were further divided into 18 sub-clusters. Of the 18 sub-clusters, 3 were urban type, 6 semi-urban and 9 rural types.

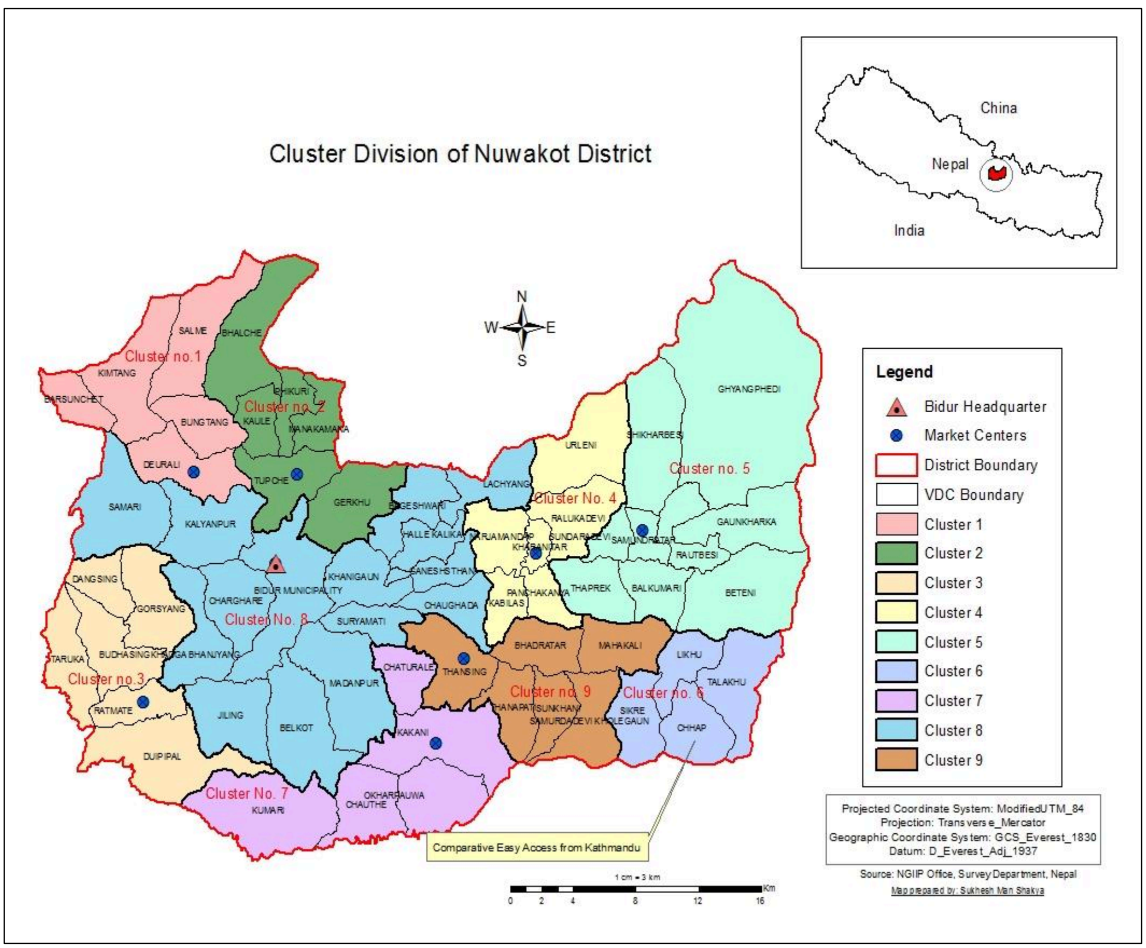


Figure 1: Cluster Map of Nuwakot

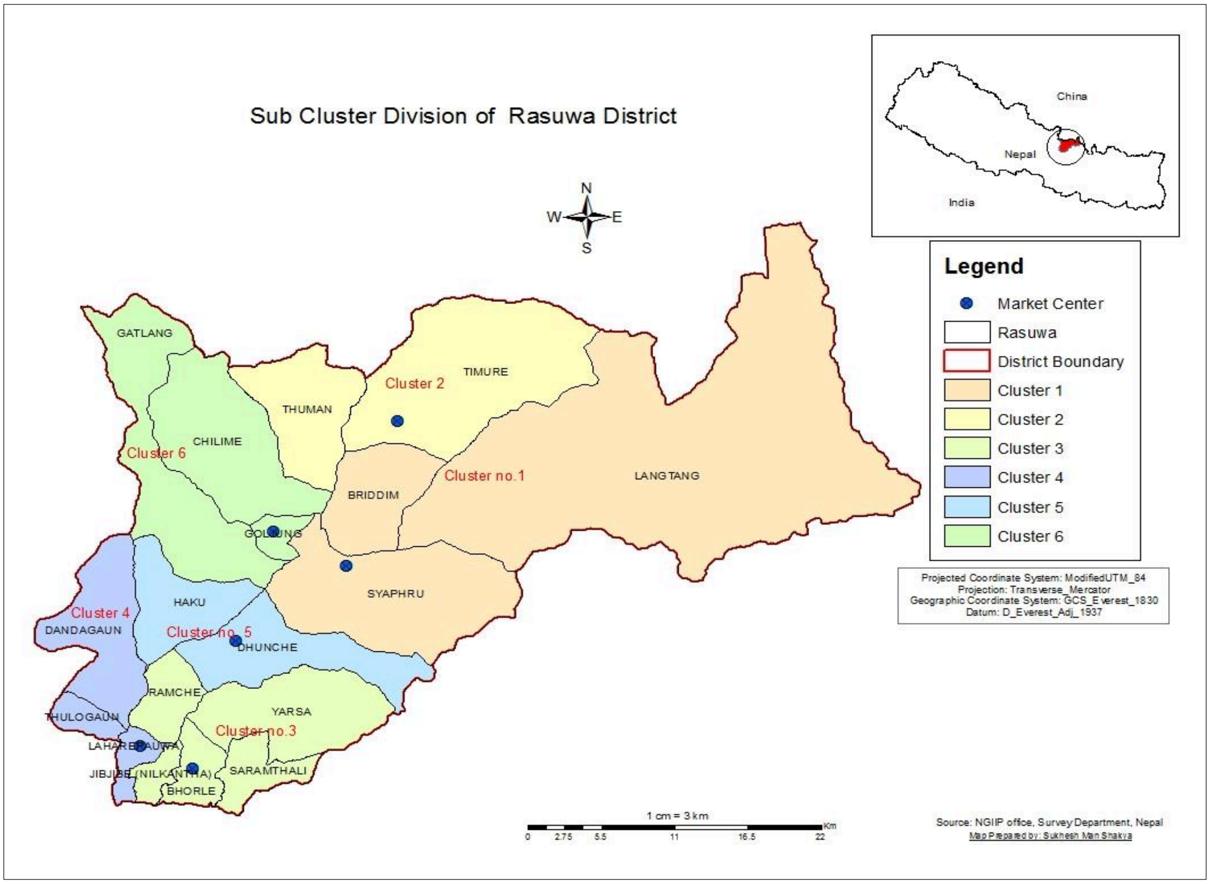


Figure 2: Cluster Map of Rasuwa

2.4 Identification of Stakeholders for Interview

It was realized that no single stakeholder can provide complete information about the problems of housing reconstruction. As a result, the study team has identified the five types of stakeholders for interview.

- VDC/NP secretaries for intending to collect overall VDC level information on road, people economic status, housing, market, natural resources etc.
- Earthquake victims for intending to collect information on their housing reconstruction plan, choice of construction materials, availability of usable salvaged materials, access to market for non-local materials, and availability of local materials etc.
- Vendors of construction materials for intending to collect information on sale, price fluctuation, sales figures, storage capacity, demand fluctuation of non-local materials etc.
- Manufacturers of local-construction materials for intending to collect information on volume of production, markets of production, difficulties faced in the production etc.
- Transportation service providers for intending to collect information on coverage of VDCs by four wheeler and transportation cost from one market to another etc.

2.5 Data Collection Methods

Face-to-face interview was possible with all types of stakeholders, except earthquake victims. An appropriate probability sample survey of earthquake victims could not be designed due to the absence of reliable VDC or ward level data on damages caused by the earthquake, which is prerequisite for any probability sampling technique. As a result, it was decided to collect information from earthquake victims through focus group discussion (FGD).

2.6 Sample Selection Method

Due to the lack of very basic information, it was not possible to select samples based on probability sampling technique. An alternative approach adopted in this study is as follows.

- Collect information of all 80 VDCs/NP from VDC secretaries
- Collect information of all 80 VDCs/NP from earthquake victims
- Collect information from at least one vendor, if available, from each cluster
- Collect information from at least one manufacturer, if available, from each cluster
- Collect information from at least one transport service provider, if available, from each cluster

2.7 Data Collection Tools

Semi-structured questionnaires were developed for collecting both quantitative and qualitative data. To ensure the reliability and validity of the tools the following tasks were carried out.

- Tools were pre-tested in and revised them as per pre-test
- Field workers had given two days rigorous training. The training included orientation on the questionnaires, role play and field practice. Besides, basic tips on FDG facilitation were provided in the training.

2.7.1 Focus Group Discussions (FGD)

Focused group discussion was carried out in each sub-cluster to collect qualitative information about the damage, materials and supply chain situation in the sub-cluster. Some quantitative information was also collected from FGD however they were only used for the validation and cross checking purpose. Mainly the following information were collected from the FDG

- Damage due to earthquake
- Choice of construction materials
- House building plan
- Market for construction materials
- Finance
- Availability of local materials

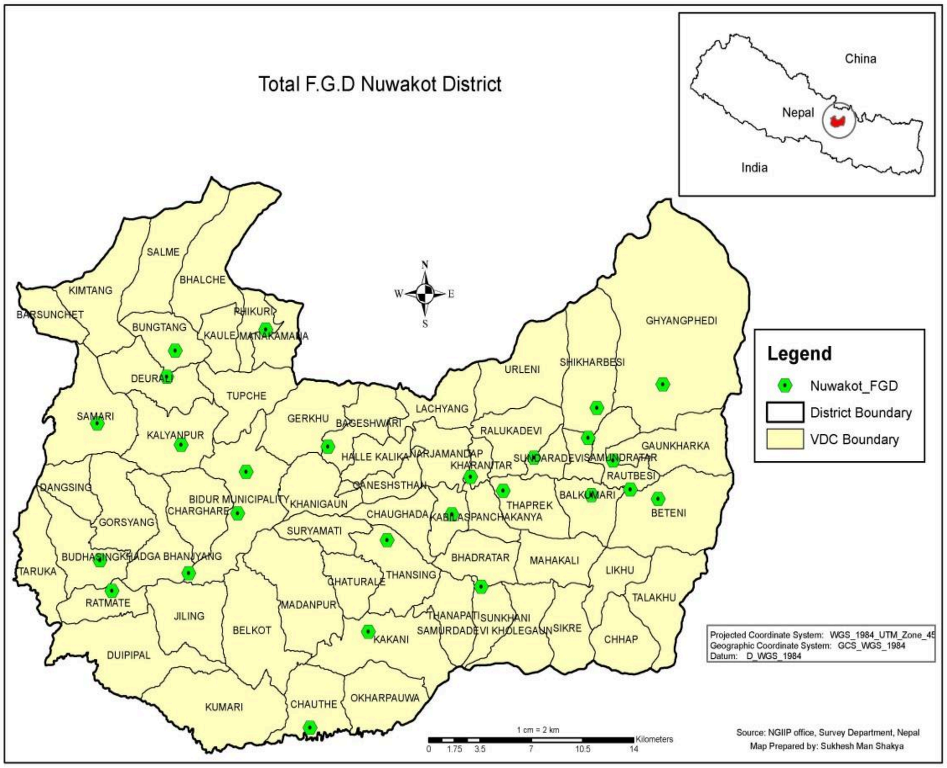


Figure 3: FGD in Nuwakot

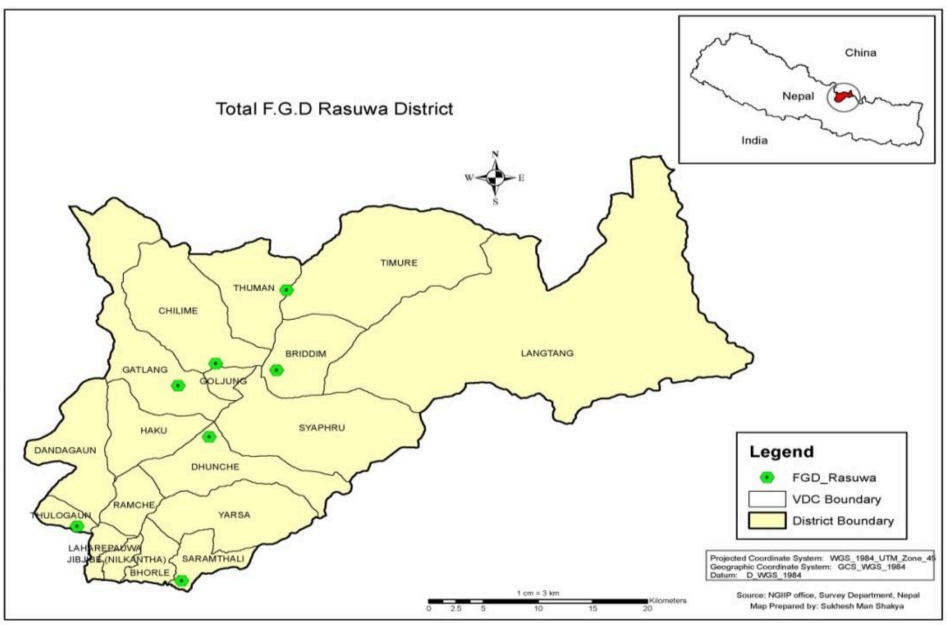


Figure 4: FGD in Rasuwa

2.7.2 Key informant interview

Key informant interview was carried out with the VDC secretaries of all the VDCs of Nuwakot and Rasuwa districts. Likewise, all the vendors and producers of the construction materials at the selected (9 in Nuwakot and 6 in Rasuwa) market centers were interviewed. Besides, selected (no of Vendors and Producers), from the regional markets like Narayangradh, Bhairawa and Birgunj were interviewed. An interview guideline and checklist (manufacturing level, vendor level, transportation level, VDC secretary level) was developed to facilitate interviews. These interviews were semi-structured to increase flexibility and develop a fuller qualitative understanding of the market actor's conditions. Quantitative data such as production, sale, price fluctuation, sales figures, storage capacity, and demand fluctuation were collected from the KII. The following table summarizes the total number of key informants interviewed

Table 1: Number and Types of Key Informant

Key informants	Nuwakot	Rasuwa	Total
VDC secretaries	55	18	73
Vendors	11	5	16
Producers	24	8	32
Transporters	8	5	13

2.7.3 Resource Mapping

Following the focused group discussion, resource map of each cluster was prepared. Selected participants of the FGD were mobilized to prepare the resource map. Mainly the natural resources like the stone quarries, rivers, streams, jungles, sand quarries were mapped in the resource map. The Resource maps were first prepared in loose brown sheet, and then the enumerators transferred them to cluster maps which the enumerators were provided with.

2.7.4 Mobility Map

Mobility map was also prepared for each cluster for tracing the mobility of people from the cluster to various market centers for the procurement of the non-local construction materials. The map mapped \ various local and regional markets where people from the cluster visit for material procurement. The map also indicates the distance of the markets from the cluster and the access condition. It also shows the relative importance of various markets with regard to material supply to the cluster.

2.7.5. GIS mapping

Spatial location of construction materials vendors and the producers at each cluster was taken with the help of Global Positioning System (GPS). The location was then transferred to the Global Information System (GIS) along with basic attributes like name of the vendor/enterprise, material sold/produced.

2.8 Method of Data Processing

In order to process the data in computer more efficiently, coding schemes were developed for all the qualitative responses. To expedite the data entry job, data entry programs were written in CSPro6.3. Finally, the descriptive type of data analyses was carried out using SPSS20.

2.9 Interaction and discussion with the stakeholders

The study team held meeting and interaction with the concerned stakeholders including NRA , on the periodic basis, throughout the study in order to accommodate their concern and update them with process and progress of the study. The meetings were also important for the validation of the survey finding. The table below summarizes the meeting held with different stakeholders.

Table 2: List of meetings conducted with different stakeholders

Stakeholders	Objective of the meeting	Date
NRA	Share the objective and assessment plan Share survey methodologies and questionnaires'/checklist Share the early impression of the field study Share the preliminary finding Share the draft report	
DDC	Share the study objective and assessment plan Share the preliminary findings	
DAO	Share the study objective and assessment plan Share the preliminary findings	
DUDBC	Share the study objective and assessment plan Share the preliminary findings	

After data analysis, the preliminary findings of the study were shared with the district stakeholders for the validation of the findings. The relevant stakeholders were invited in the workshop (list of the stakeholders participated in the workshop at the annex). Likewise, the preliminary findings were shared with NRA for the feedback and comments. The final report incorporated the feedbacks and comments.

2.10 Study Limitation

The followings are the limitations of the study:

- The study resorted to focus group discussion (FGD) to collect information at cluster level and use it for drawing inference. As cluster represents a large geographic area with 3-5 VDCs, the accuracy of generalization of conclusion depends on accurate representation of respondents and information drawing technique.
- The clusters were divided into urban, semi urban and rural according to their access to different types of road. As mentioned earlier, VDCs having the all-weather road was categorized as the Urban, VDCs with the gravel road as Semi Urban and VDCs with only fair weather road or no road as rural. However, the settlements with in a VDC also have different level of access. For instance, some settlements in a VDC categorized as Urban may have only the earthen road. Hence, the categorization may not fully represent the ground reality.
- This study is not based on household survey of victims. Consequently, it does not show the socio-demographic and economic conditions of the victims which limit to identify who are the most vulnerable victims that may need special packages during the reconstruction phase.
- Some qualitative data in this study are just the perceptions or opinions of the respondents, and similarly some quantitative data are just the intelligent guesses of the respondents. In this context, extreme caution should be taken while drawing definitive conclusions.
- There were also some tendencies to exaggerate and distort the information. Likewise, the information received from the regional level vendors /manufacturer may not be complete as despite of our attempts to clarify our objective of the study, some of them were still skeptical about the use of the data

3. Study Findings and discussion

The findings of this study are based on analysis of both quantitative and qualitative data collected from 54 VDCs and 1 Municipality of Nuwakot and 18 VDCs of Rasuwa.

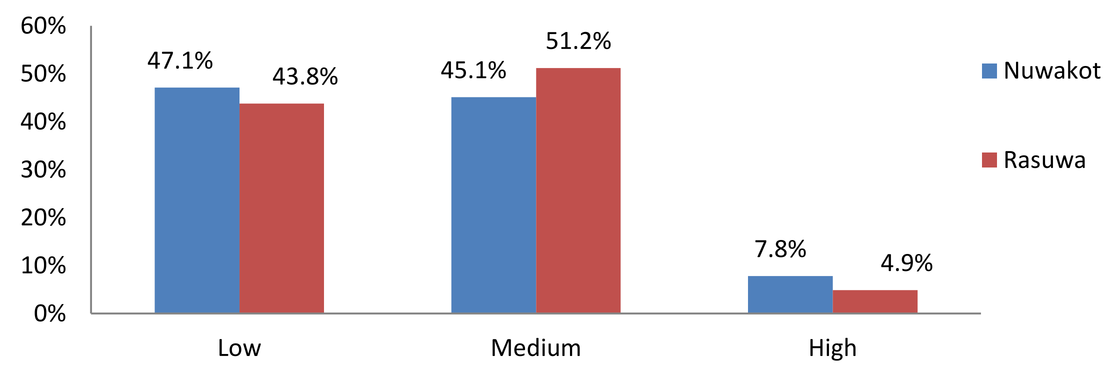
3.1 Situation assessment

The economic status of the communities and their access to road network has direct bearing on the peoples' choice of construction materials. Hence, it was important to assess the socio-economic status of people in the study districts. However, as collection of the quantitative information on socio-economic status would constitute a separate comprehensive study on its own right, qualitative information were collected from the VDC secretaries.

3.1.1 Socio economic status:

A clear pattern of the economic status is observed in both Nuwakot and Rasuwa districts. Around 43-47 per cent people are under low income status while around 45-51 per cent people are under medium income group. Most of the poor households come from rural areas with less accessible road network. The severity of damage is concentrated in the areas where people with low income status are dominant.

Figure 5: Economic status of people



3.1.2 Access to road network:

All the VDCs (54 VDCs) and almost all the wards (486 wards) in the VDCs are connected with Road network in Nuwakot district. Only about 3.6 % of the wards (17 wards) don't have the road access. However, only 9.4% of the total wards (45 wards) in the district are connected with blacktopped road. Of the total wards connected with blacktopped road, majority lies in the urban areas. Only negligible number of wards in rural areas (1.4%) are connected with blacktopped road. 85.4% of the wards in the rural areas of Nuwakot have access to only earthen roads which become impassable during the rainy season.

Contrary to Nuwakot, 42% of the wards in Rasuwa are not connected with any form of road. Only 8.6 % wards are connected with black topped road, of which majoring lies in urban areas. Majority of the wards have only the earthen road, which are operational only few months of dry season.

Figure 6: Road Network of Nuwakot

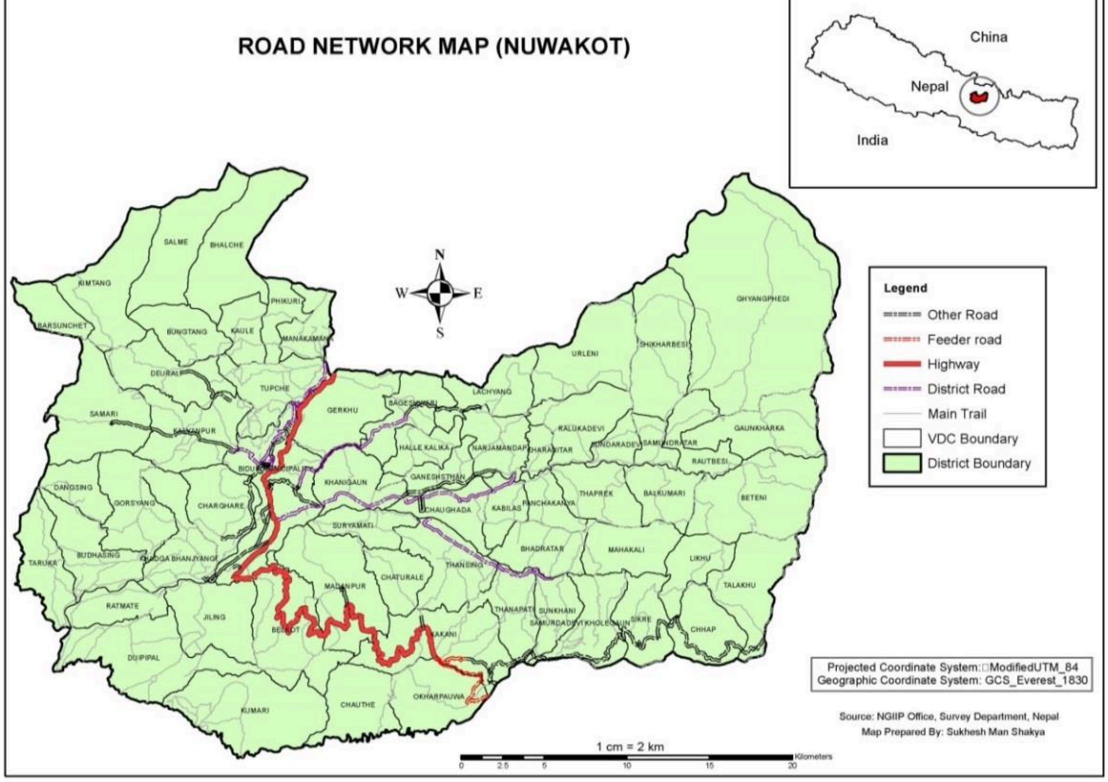


Figure 7: Road Network of Rasuwa

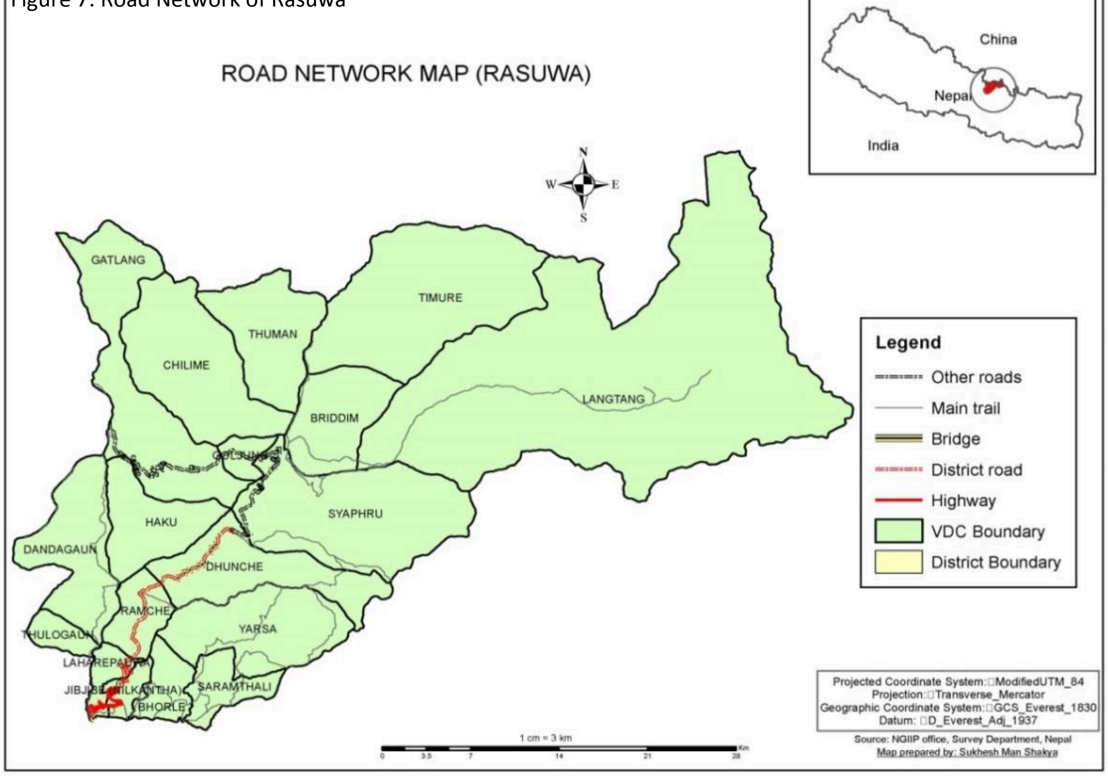


Table 3: Percentage of wards by road type within each sub-cluster

	Urban type	Semi-urban type	Rural type	Overall
Nuwakot				
BT	44.7%	6.7%	1.4%	9.4%
GR	19.6%	15.5%	2.8%	12.3%
ER	35.7%	76.8%	85.4%	74.6%
None	0.0%	1.0%	10.4%	3.6%
Total	**100.0%**	**100.0%**	**100.0%**	**100.0%**
N	**56**	**297**	**144**	**497**
Rasuwa				
BT	37.0%	7.4%	0.0%	8.6%
GR	11.1%	1.9%	0.0%	2.5%
ER	51.9%	55.6%	39.5%	46.9%
None	0.0%	35.2%	60.5%	42.0%
Total	**100.0%**	**100.0%**	**100.0%**	**100.0%**
N	**27**	**54**	**81**	**162**

Note: BT- Black topped; GR- Gravel road; ER- Earthen road

3.2 Damage assessment

3.2.1 Damage assessment at ward level

Both Nuwakot and Rasuwa districts are among the 7 most severely hit districts. However, the severity of the damage is not uniform across the districts. Here, the severity refers to the number of casualties and collapsed houses rather than the damaged houses.

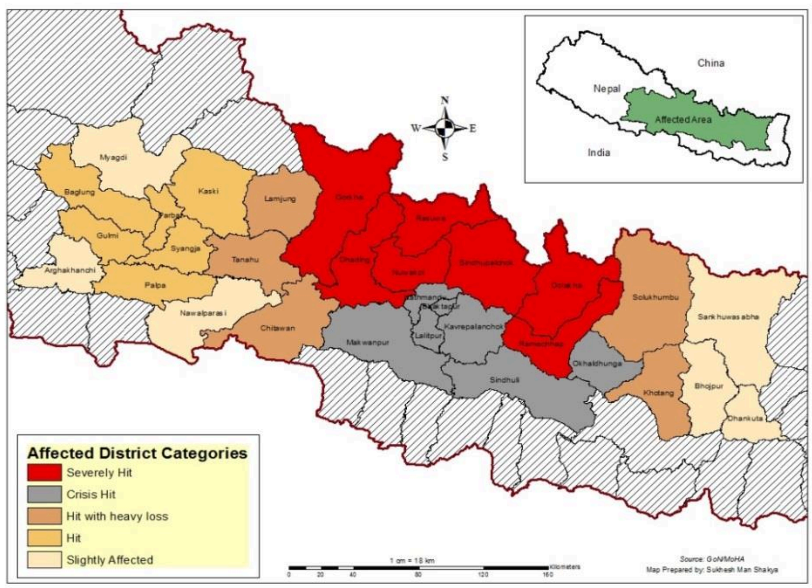


Figure 8: Affected Districts Category

The severity of damage is higher in semi urban areas of Nuwakot. Around 85% of wards under semi-urban VDCs are severely damaged by earthquake. Similarly, around 80% of wards under rural VDCs are severely damaged. The damage is least severe in wards under urban VDCs with only 35.7% severe damage.

In case of Rasuwa, the severity is more uniform across all geographic areas. Around 90% of wards under urban VDCs are severely damaged, while the percentage for severe damage under semi-urban and rural wards is around 77% and 80%.

Table 4: Percentage of wards by extent of damages within each sub-cluster

	Urban type	Semi-urban type	Rural type	Overall
Nuwakot				
LOW	16.1%	3.0%	0.0%	3.6%
MEDIUM	48.2%	12.1%	20.1%	18.5%
HIGH	35.7%	84.8%	79.9%	77.9%
Total	100.0%	100.0%	100.0%	100.0%
N	**56**	**297**	**144**	**497**
Rasuwa				
LOW	3.7%	7.4%	0.0%	3.1%
MEDIUM	0.0%	14.8%	19.8%	14.8%
HIGH	96.3%	77.8%	80.2%	82.1%
Total	100.0%	100.0%	100.0%	100.0%
N	**27**	**54**	**81**	**162**

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Figure 9: HH Damage Map Rasuwa

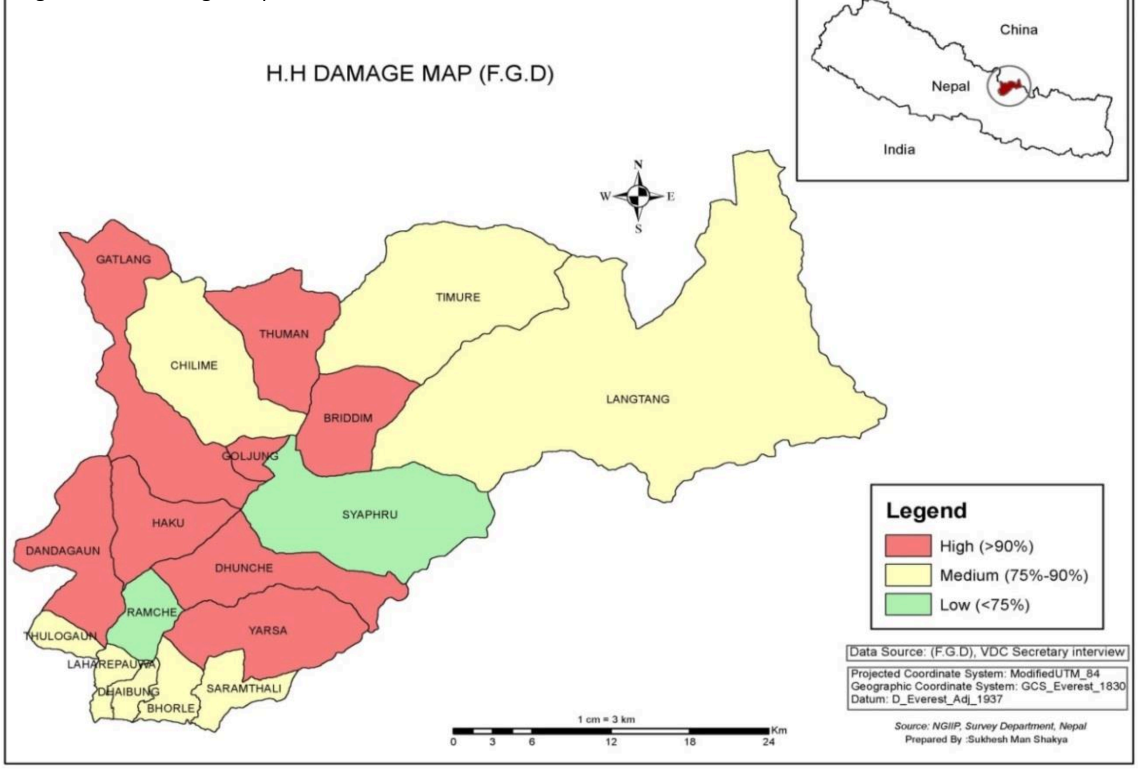
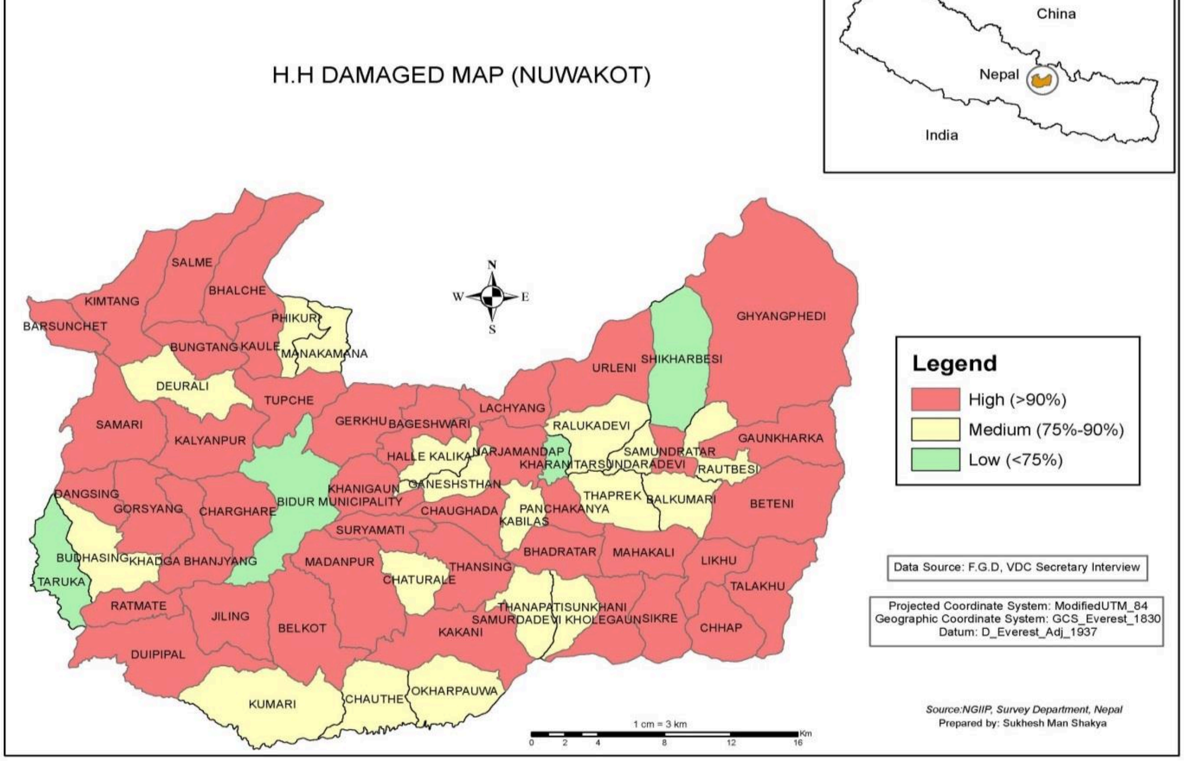


Fig 10: HH damage map Nuwakot



3.2.2 Damaged houses by different clusters

The analysis of damages by cluster indicates that cluster 2 (comprises of VDCs: Bhalche, Kaule, phikuri, Manakamana, Tupche and gerkhu) received maximum damage. A total of 26551 houses in cluster 8 have damaged due to earthquake which is around 36 per cent of total number of house damaged in Nuwakot (72857 houses damaged). Similarly in case of Rasuwa, cluster 1 (comprises of VDCs: Bridhin, Langtang, Syaphru) received maximum damage. A total of 5436 houses damaged in cluster 3 which is equivalent to 44% of total house damaged in Rasuwa districts (12116 huseses damaged)

Figure 10: Damage houses by cluster in Nuwakot

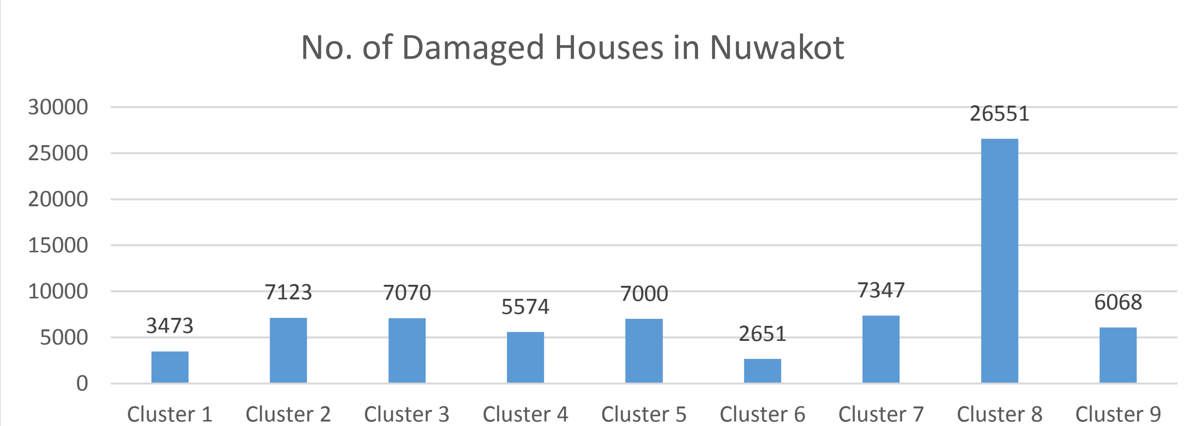
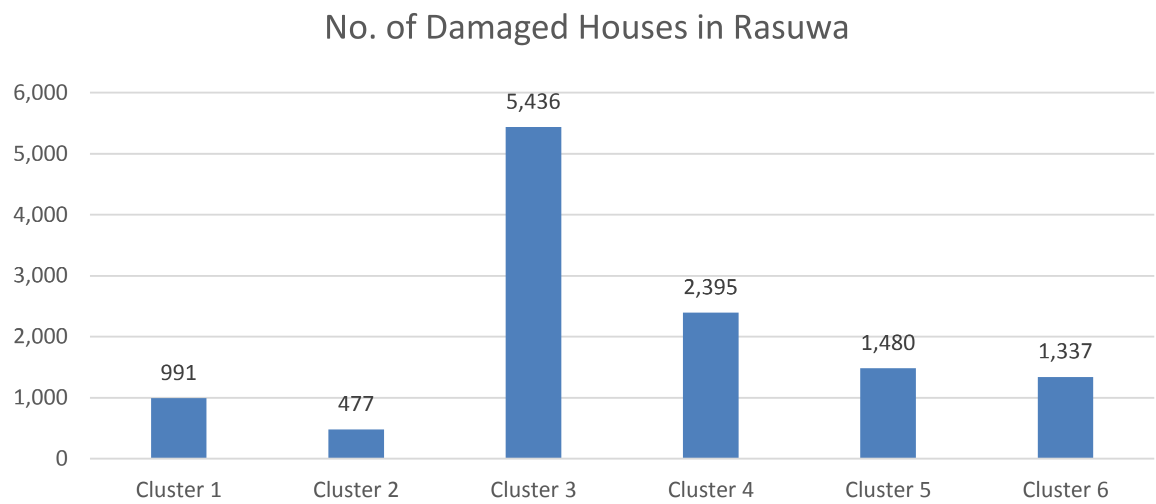


Fig 12: Damage houses by cluser in Rasuwa



3.2.3 Number of damaged houses by different sub-clusters

In Nuwakot district, 39728 houses damaged lies in semi urban area which is around 55% of total houses damaged. Number of houses damaged in rural area is 18537 (25%) while it is 14592 (20%) in urban areas. Contrary to this, in Rasuwa districts, damaged houses in rural areas shares around 43% of total houses damaged in the district.

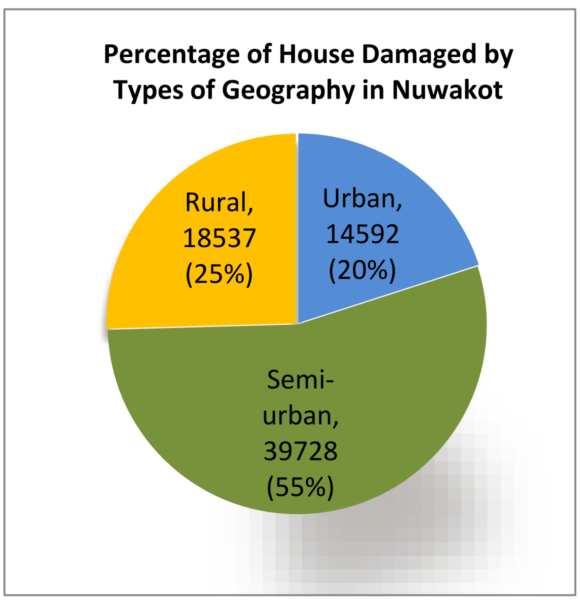


Figure 11: Damage by geography in Nuwakot

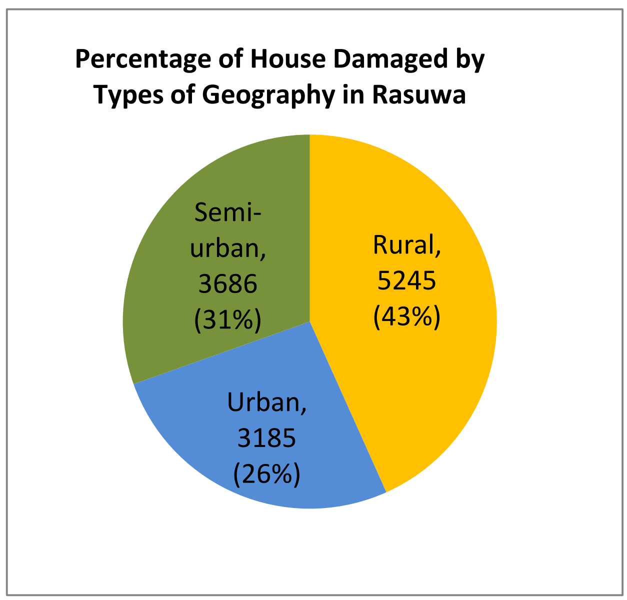


Figure 12: Damage by geography in Rasuwa

3.3 Demand Situation

3.3.1 Building construction type before and after earthquake

Before the earthquake almost all the houses (89% in Nuwakot and 76% in Rasuwa) were stone masonry mostly in mud mortar. However, following the earthquake, people are apprehensive about stone masonry. As they have witnessed more stone masonry buildings collapsed, they have developed a false notion that stone is inferior to brick as the construction material. Hence, people have more preference towards the brick masonry. However, the brick masonry is beyond the scope of majority of the people in the study areas. As a result, majority of people have no option but go for the stone masonry. 67 % of people in Nuwakot and 74% of people in Rasuwa are expected to build the stone masonry building during the reconstruction. The figure can go still higher if people's misunderstanding about stone masonry is clarified. Of the total planned stone masonry buildings, 11.8% and 23.6% in Nuwakot and Rasuwa districts respectively are likely to use cement mortar. Previously only 1.6% of stone masonry buildings in Nuwakot and 6.6% of the building in Rasuwa were in cement mortar. Likewise, significant numbers of households in Nuwakot district are planning to build the brick masonry buildings. The figure stands at 26.4% in Nuwakot, which is a remarkable increase from the pre-earthquake figure of 7.8%. Likewise, number of households planning to build brick masonry building is more than double in Rasuwa district. Previously, only 3.4% of the houses were brick masonry buildings. People intending to use other construction materials are 4.7% in Nuwakot and 15.7 % in Rasuwa district. However, this may also increase, once people have more awareness about the alternative construction materials as the local materials are scare and the non-local materials are too expensive for many households.

Table 5: Percentage of houses by type before the earthquake reported by secretaries

Sub-cluster	Mud bonded stone	Mud bonded brick	Cement bonded stone	Cement bonded brick	Others	N
Nuwakot						
Urban type	76.1	2.5	2.5	16.5	2.5	**6**
Semi-urban type	89.4	1.9	1.7	6.1	0.9	**33**
Rural type	97.8	0.0	0.4	1.6	0.2	**16**
Overall	**87.9**	**1.6**	**1.6**	**7.8**	**1.1**	**55**
Rasuwa						
Urban type	72.8	0.0	7.2	5.0	15.0	**3**
Semi-urban type	63.9	2.7	11.4	3.4	18.6	**6**
Rural type	72.9	0.0	2.2	2.3	22.6	**9**
Overall	**69.9**	**0.9**	**6.6**	**3.4**	**19.1**	**18**

Table 6: Percentage of households intending to build houses by type within each sub-cluster

	Mud bonded stone	Mud bonded brick	Cement bonded stone	Cement bonded brick	Other	N
Nuwakot						
Urban type	29.1	2.6	11.9	51.6	4.7	**6**
Semi-urban type	57.0	2.2	13.6	22.2	5.0	**33**
Rural type	79.2	0.4	6.9	9.8	3.6	**16**
Total	**55.2**	**1.9**	**11.8**	**26.4**	**4.7**	**55**
Rasuwa						
Urban type	55.0	0.0	14.9	11.8	18.3	**3**
Semi-urban type	35.2	0.9	39.2	13.7	11.0	**6**
Rural type	60.2	0.0	16.7	5.2	17.8	**9**
Total	**50.6**	**0.3**	**23.6**	**9.9**	**15.7**	**18**

With regard to the roofing materials, overwhelming 82% of households in Nuwakot and 73.3 % of households in Rasuwa prefers CGI sheet as a roofing material. CGI roofs are equally popular in urban, semi urban and rural areas. The CGI sheet which was the most popular roofing materials even before the earthquakes is all set to replace almost all other roofing materials during the reconstruction. The use of slate is likely to plunge drastically during reconstruction. Previously, 16.2% and 9.6% of houses had the slate roofs in Nuwakot and Rasuwa districts respectively. However, Only 1.7% of people in Nuwakot and 4.4 % of people in Rasuwa are planning to use slates during reconstruction. The slates are heavier than the CGI sheets. Hence, they not only make the super-structure heavier but also require heavier wooden rafters. This increases the wood requirement. Likewise, fixing the slate requires more skill compare to fixing the CGI sheet. Besides, the slates are not locally available as popularly assumed. Hence, people have developed distaste towards the slates.

To the sharp contract to the slates, a large number of household is planning to go for the RRC roofing during the reconstruction. Before, only 3.8% of houses in Nuwakot and 4.0% of houses in Rasuwa had the RCC Roof. About 16% of households in Nuwakot and 22.3 % of houses in Rasuwa are planning to have the RCC roof during reconstruction.

. Table 7: Residents intending to build houses by roof type within each sub-cluster

	CGI Sheet	**RCC**	**Slate**	**Others**
Nuwakot				
Urban Type	77.7	17.2	4.4	0.7
Semi-Urban Type	83.5	16.0	0.5	0.0
Rural Type	82.0	14.8	1.8	1.4
Overall	**81.8**	**16.0**	**1.7**	**0.5**
Rasuwa				
Urban Type	77.5	22.5	0.0	0.0
Semi-Urban Type	76.9	23.1	0.0	0.0
Rural Type	67.1	21.6	11.3	0.0
Overall	**73.3**	**22.3**	**4.4**	**0.0**

3.3.2 Demand for major construction materials

Demand for local materials: 84973 houses need to be rebuilt during reconstruction phase in Nuwakot and Rasuwa districts. Hence, the demand for the construction materials is massive. Table 9 presents the approximate quantity of local materials required for the reconstruction. Among the construction materials the demand of stone is huge as majority of the people in both the districts still opt for the stone masonry. If we add on it the demand for aggregate, which also requires stone, the demand for stone further increases.

Table 8: Demand of local construction materials

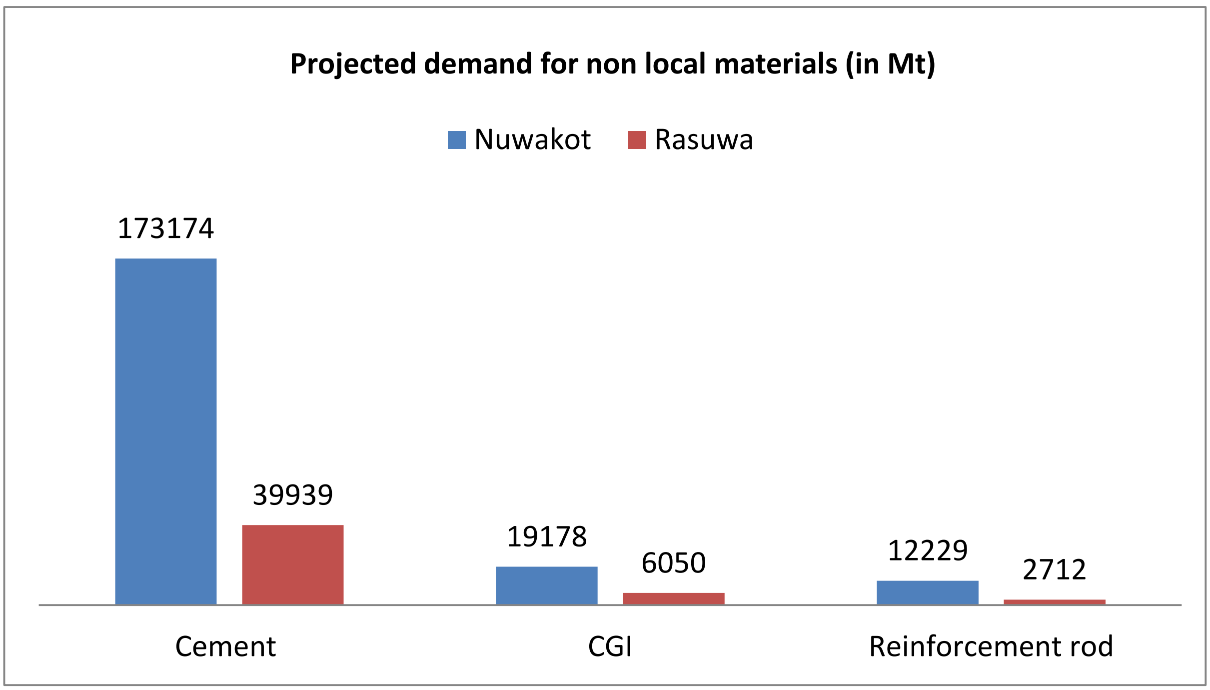
Districts	Category	**Stone (M3)**	**Mud (m3)**	**Sal Wood (m3)**	**Sand (m3)**	**Aggregate (m3)**
Nuwakot	Rural	850959.52	589847.34	119772.93	55388.56	27323.53
	Semi Urban	1429175.07	936388.96	209128.19	251160.4	124587.00
	Urban	500447.23	334302.72	83977.98	151143.9	77600.25
	Total	**2780581.83**	**1860539.02**	**412879.10**	**457692.9**	**229510.80**
Rasuwa	Rural	850959.52	2635.18	119772.93	55388.56	27323.53
	Semi Urban	120487.96	53059.97	14493.27	39362.79	18488.97
	Urban	112481.46	70070	15348.80	16077.88	7752.29
	Total	**1083928.95**	**125765.15**	**149615.01**	**110829.2**	**53564.80**

Demand for Non local materials: The table 10 summarizes the demand for various non local materials. The government approved house designs (SMM 1.1, SMC 2.1) were considered for the estimation of the non-local materials. Irrespective of the urban, semi urban and rural areas, people prefer CGI sheets as the roofing materials. Hence, there is huge demand for CGI sheet. Besides, CGI Sheets, there is significant demand for other non-local materials like cement, brick and reinforcement. The demand for the materials is highest in semi–urban areas of Nuwakot and Rural areas of Rasuwa as maximum number of households in the districts falls under the category.

Table 9: Demand of non-local construction materials

Districts	Category	Non local materials				
		CGI Sheet (Bundle)	Plain Sheet (R.m)	Cement (Bags)	Reinforcement (Kg)	Brick (Numbers)
Nuwakot	Rural	74500.203	147554.52	410186.7	1424012.34	40532411.02
	Semi Urban	165189.024	235189.76	1873652	6542407.04	208871549.1
	Urban	79949.568	84049.92	1179646	4262323.2	169691827.2
	Total	**319638.795**	**466794.2**	**3463485**	**12228742.6**	**419095787.3**
Rasuwa	Rural	74500.203	147554.52	410186.7	1424012.34	40532411.02
	Semi Urban	15072.054	13306.46	273069.9	896951.24	11562561.8
	Urban	11258.975	17517.5	115513.6	391181.7	8024722.16
	Total	**100831.232**	**178378.48**	**798770.3**	**2712145.28**	**60119694.97**

Figure 13: Projected demand of non-local materials



3.4 Supply Situation

3.4.1 Supply situation o f local construction materials

Traditionally, the rural households in Nepal manage to get the local construction materials by themselves from the natural resources available in their vicinity. With the natural resources depleting and various restrictions imposed on exploitation of the natural resources, it has become increasingly difficult to manage all the local materials by themselves (without any cost). During focused group discussion (FGD) many of the participants complained that local materials like sand, stone and woods are no more local materials for them as they have to purchase it from outside. The study, assessed, how much of the local materials can still be managed locally. The study has also identified the major sources of the materials. The materials wise results are discussed below.

***Stone*:** In terms of availability of stone, Rasuwa district is in a comfortable position. About 97.8% of the demand for stone can be met locally in the district. There is 13% deficit in urban areas. However, considering the small number of households in urban areas, the gap is manageable. Alternatively, Nuwakot district is in a less comfortable position in this regard. Overall, only 73.9% of the demand for stone can be met locally in the district. The situation is worse in urban areas where only 52.4% of the demand can be met locally. The salvaged stones from the damaged houses are the major source of stone. It is estimated that 89.1 % and 87.5 % of the stone could be salvaged from the ruined buildings. The stones such recovered can meet about 45 to 50% of the demand as the stone requirement for the government approved building is more than the traditional houses. Likewise, all the stone such recovered cannot be used in the new building. The households are planning to manage the remaining demand, from own land, local stone quarries and rivers. The following table summarizes, the weight of different sources in terms of households dependency on the sources for stone

Table 10: People dependency on different sources for stone

District	Own land	Stone quarry	River
Nuwakot	59%	52%	46%
Rasuwa	61%	78%	6%

Note: Total percentage more than 100%, represent that people are planning to collect the stone from more than once source to meet their reconstruction demand.

Timber: Slightly over fifty percent**,** 55.2 % and 52.9 %, respectively in Nuwakot and Rasuwa districts of the total demand for wood can be met locally. Less wood are locally available in the urban and semi urban areas of the districts, resulting about 65% and 43% deficit respectively in Nuwakot and Rasuwa districts. The semi urban and rural areas are slightly better off in this regard, with more than 50% of demand can be met locally at Nuwakot and 60% in Rasuwa.

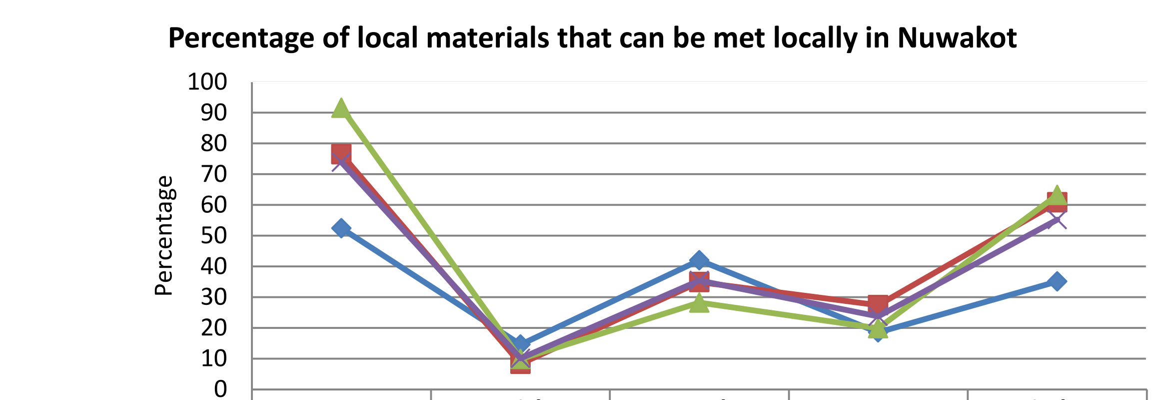
The salvaged wood from the damaged building is the major source for wood as well. It is estimated that 49.1% and 68.8% of the wood can be salvaged from the damaged building. However, the salvaged wood can cover only about 25 to 30% of the requirement of the new building, as the government approved design requires more wood and all the wood salvaged couldn't be used in new construction. Private forest (trees grown in the private land) is the primary source of timber for 93 % households in Nuwakot and 56% of households in Rasuwa. 80% of households in Nuwakot and 70% of households in Rasuwa also access wood from community forest. considering the limited forest resources and the restriction imposed on harvesting timber from community forest and National parks (% covered by forest and nationa parks , meeting the gap is really challenging.

Brick: Around 10% of total demand for brick can be met by local production of brick in Nuwakot district while in case of Rasuwa it has to depend upon import of brick from other districts particularly from terai areas. The demand for brick is higher in urban and semi-urban areas where many people intend to replace brick with stone, so the demand will be quite higher in urban and peri urban areas.

Sand*:* Around 35% of total demand of sand in Nuwakot district can be met from local production while in case of Rasuwa it is only around 30%. Due to existing legal provision to limit sand extraction to specified quantity during specific months of the year limits the production capacity of sand entrepreneurs. If proper measure is not taken to increase supply of sand, there is a high possibility to increase the price of sand during reconstruction.

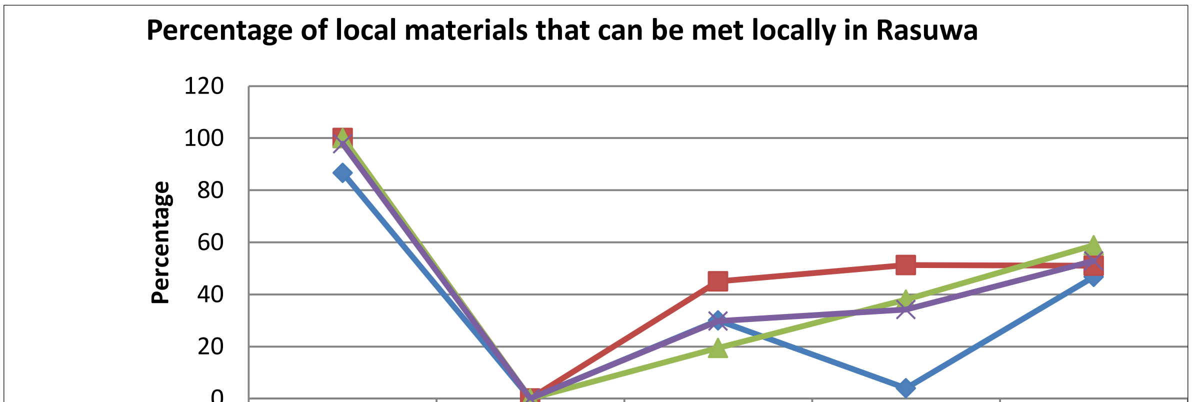
Aggregates: The supply of aggregates from local source can hardly meet the demand of aggregates during reconstruction. Only 23% of total demand can be met in Nuwakot while the case of Rasuwa is little bit better off with 34%. This indicates a high opportunity to promote aggregate enterprise in both districts. More challenging is to meet the demand for aggregate and sand, if the districts continue to depend on rivers only.

Figure 14: Percantage of local materials that can be met locally in Nuwakot



	Stone	Brick	Sand	Aggregates	Timber
Urban	52.4	14.5	42	18.5	35.1
Semi-urban	76.5	8.3	34.9	27.4	60.8
Rural	91.6	9.9	28.2	19.9	63.2
Overall	73.9	10.1	35.4	23.7	55.2

Figure 15: Percantage of local materials that can be met locally in Rasuwa



	Stone	Brick	Sand	Aggregates	Timber
Urban	86.7	0	30	4	46.7
Semi-urban	100	0	45	51.28	51
Rural	100	0	19.44	37.96	58.9
Overall	97.8	0	29.72	34.26	52.9

The figure below depicts the overall demand and supply situation of local construction materials in the study districts.

Figure 16: Demand and supply situationof local construction materials in Nuwakot

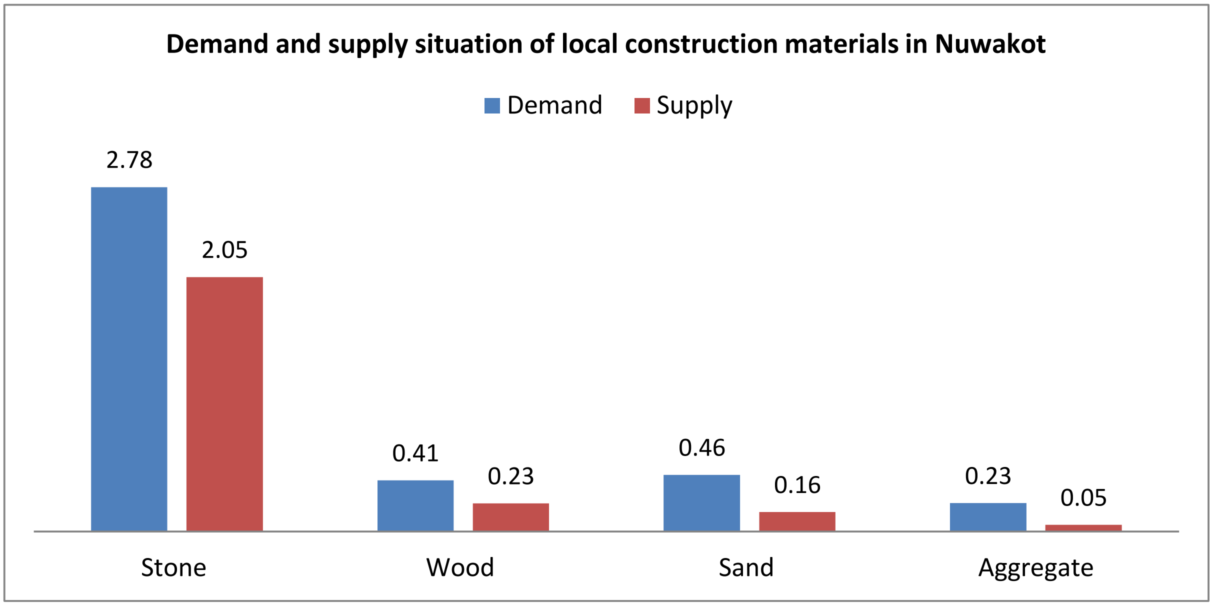
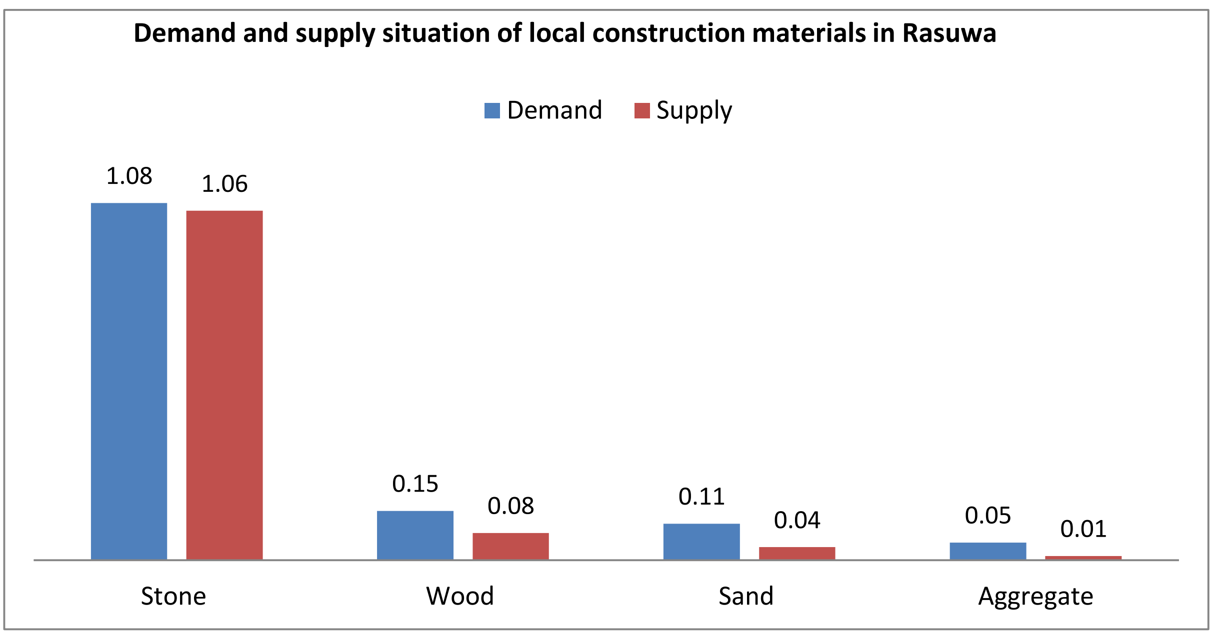


Figure 19: Demand and supply situationof local construction materials in Rasuwa



3.4.1.1 Natural resource mapping

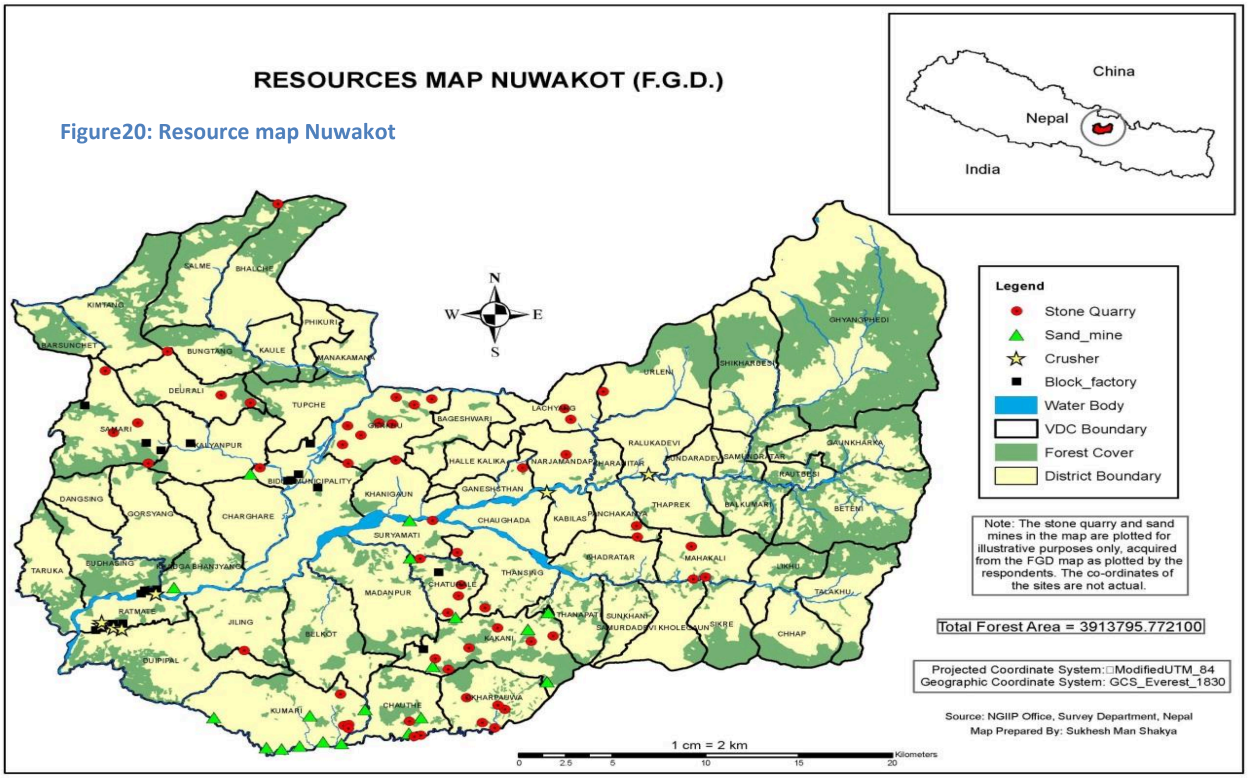


Figure20: Resource map Nuwakot

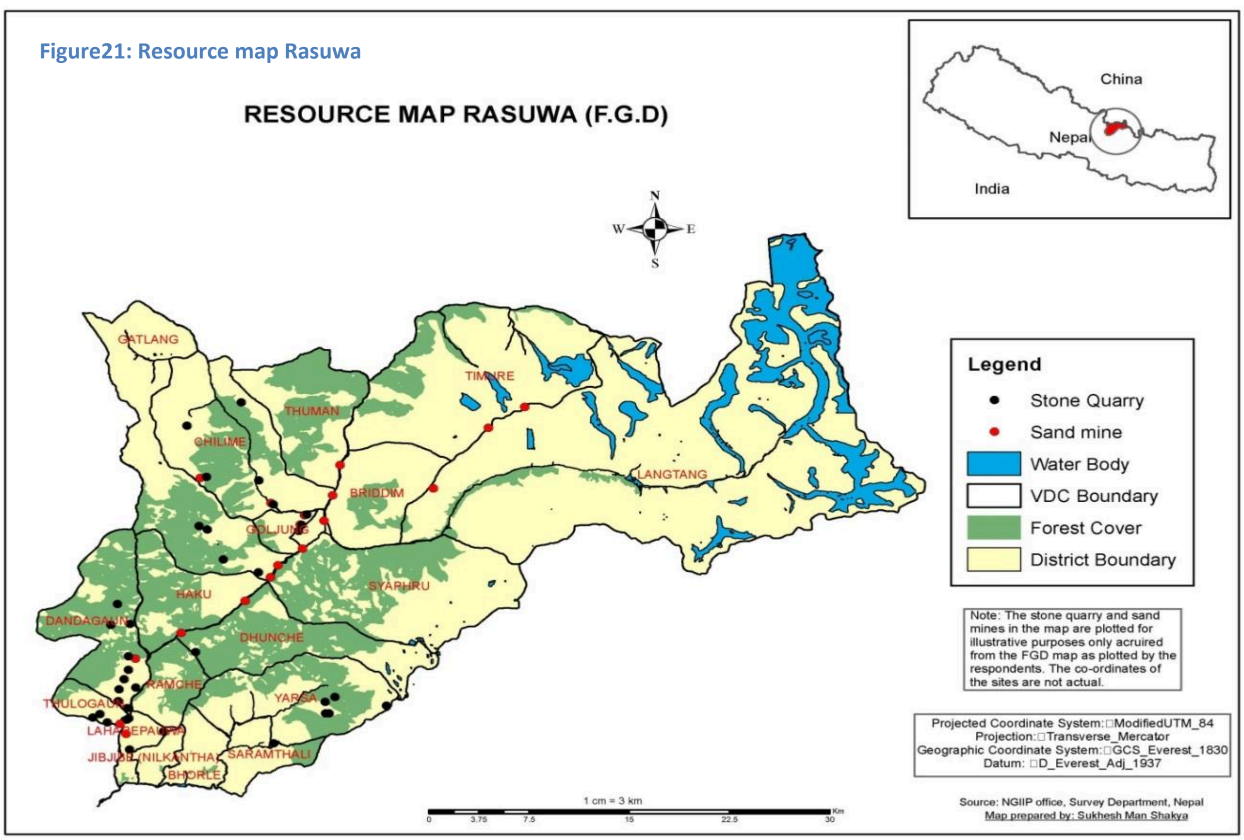


Figure21: Resource map Rasuwa

3.4.1.2 Mapping of producers of local construction materials

There found 13 brick kilns, 13 hallow concrete blocks and 8 Aggregate crushers/Sand quarries in the Nuwakot district. The combined production capacity of the producers per annum is presented in the table below.

Table 11: Supply capacity of local materials at the district level

Materials	Unit	Quantity
Sand	m^3	5360
Aggregate	m^3	4800
Brick	Number	30180000
Block	Block	2142000

Figure 172 Nuwakot Producers Map

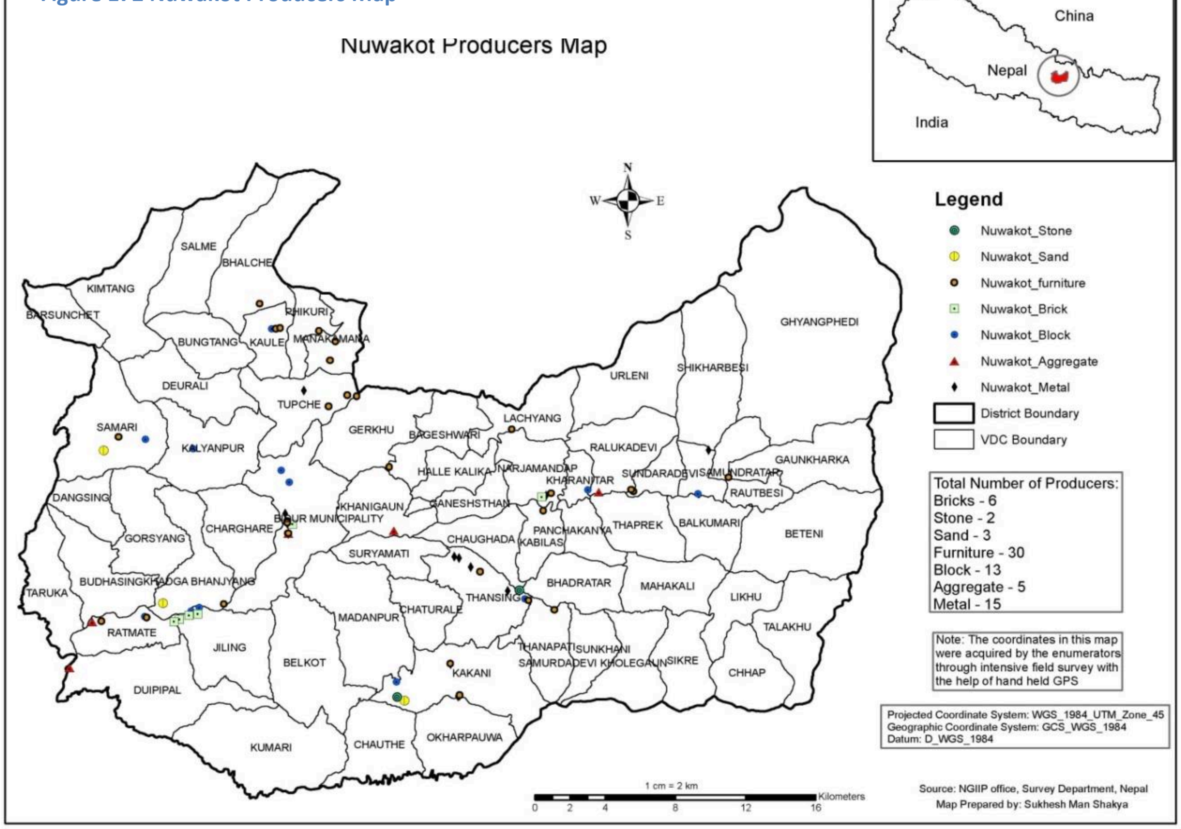
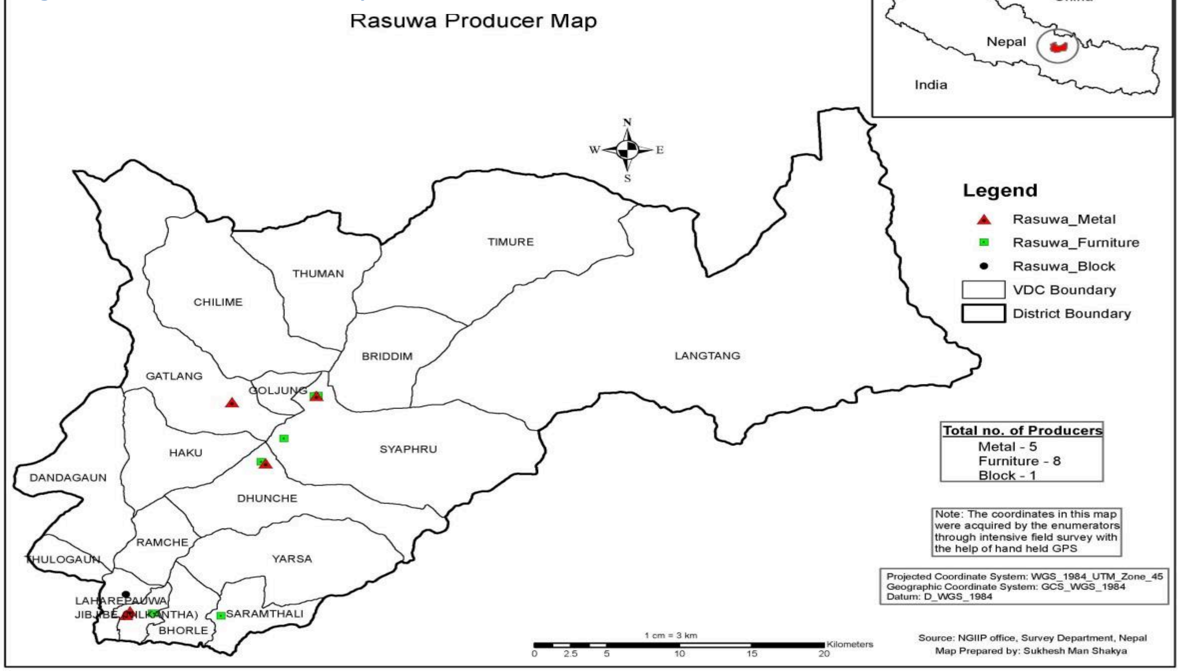


Figure 183: Rasuwa Producers Map



3.4.2 Supply situation of non-local materials

3.4.2.1 Current supply capacity of non-local materials at national level

As the supply of the non-local materials at the study districts is directly affected by the demand and supply situation at the national level, it was important to understand the demand and supply situation at the national level. Hence, the study assessed the total production of the materials at the national level and total likely demand for the materials during the reconstruction phase. Demand for the materials in 2014, was considered as the normal demand for the materials, which is assumed to continue during the reconstruction period. The normal demand plus the demand for the reconstruction is considered as the total demand during reconstruction.

Table 12: Current production of non-local construction materials at national level

Product	No. Of Factories	Production	
		Total Capacity (per day)	Current (per day)
Cement	46	19100 MT	11500 MT
Iron and Steel	16	3100 MT	2075 MT
CGI	7	1000 MT	950MT
Bricks	850	12.2 million pcs	7.5 million pcs

There are about 850 Brick Klins (ICIMOD) in Nepal with the production capacity ranging from 15,000 to 50,000 bricks per day (*Health Research and Social Development forum*). The combined production capacity of the brick Klins is 12.2 million pieces of bricks. However, after the earthquake, the production of the bricks have fallen sharply as the earthquake damaged majority of brick klins in Kathmandu valley which produce largest amount of bricks. Just after the earthquake, the brick production has plummeted to 95%. Hence, it is really challenging to meet the reconstruction demand for bricks which is roughly 4.2 billon pieces of bricks. Even if brick production takes place at the fully capacity of Klins, it will take at least 4 years to meet the reconstruction demand, if we assume that the regular demand of 3.175 billion pieces of bricks.

Similarly, there are 46 cement factories in the country with the installed capacity 19100 MT per day. However, the factories are operating only at the 60% of the capacity due to power shortage. Hence, the current production is only about 11500MT per day. Nepal imports 0.8 million ton of cement from India annually to meet the regular demand of 5 million ton. However, if the factories operate to their installed capacity, 100% of the demand for cement can be met in one year.

There are 16 steel mills in the country. If they operate in their fully capacity, they can produce 3100 MT of steel per day, which is sufficient to meet the reconstruction demand in one year. However, the current production is only 2075 MT per day, which is not sufficient to meet even the regular demand of about 0.85 million MT.
Likewise, there are 7 CGI sheet factories in the countries. The total installed production capacity of the factories is about 100MT daily. However, currently they are producing about 950 MT daily. The CGI is the only material, whose production surpasses the regular demand. Even with the current level of production the reconstruction demand can be met with in a year. However, about 50% of the CGI sheets are exported to India, resulting deficit of the CGI sheet in the country. Government has banned export of CGI sheet to India after the earthquake.

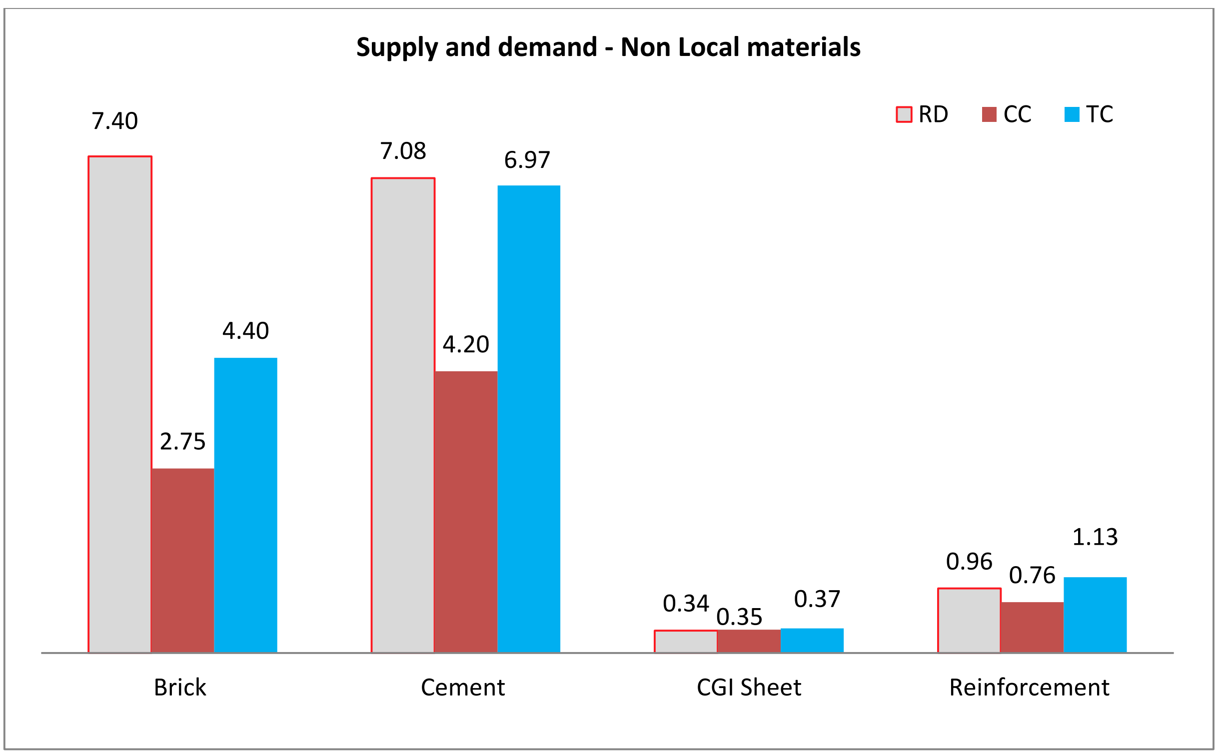


Figure 194: Supply and Demand Non-Local materials

Note: RD- Reconstruction demand; CC- current production capacity; TC- Total production capacity

3.4.2.2 Supply Chain of Major Construction materials

Cement:
Majority of the manufacturers of cement are located in Bhairahawa and Simra regions of Nepal. These manufacturers sell majority (60%) of their production to big hardware shops (regional suppliers) which are mostly located in Narayanghat, Kathmandu and Banepa. They also supply 25% of their production to medium hardware shops (district suppliers) which are present across all district of the country. The remaining 15% of their production are supplied directly to by big construction companies like hydro power, road construction etc. AT the manufacture level the average ex-factory price of PPC cement is NPR 630 and OPC cement is NPR 750.
At the regional supplier's level, approximately 50% of the goods (cement) are sold directly to consumers and the remaining 50% i.e. 30% and 20% is sold to district suppliers and local suppliers (small hardware shops) respectively. Price of PPC cement is NPR 670 and OPC cement is NPR 800 at the regional level. This variation in price from the manufacturer's level to regional level is due to transportation cost and margin of the regional suppliers.
Majority of the cement (70%) at the district supplier's level are sold directly to consumers and the remaining 30% are sold to local suppliers which are present at the VDC level. The price of PPC cement at this level is NPR 720 and OPC cement is NPR 850.
At the VDC level (where all local suppliers are present), all the goods i.e. 100% are sold directly to consumers. The price of cement is much higher here as these VDCs are remotely accessible in most cases as a result the transportation cost is significantly very high. Price of PPC and OPC cement are NPR 800 and NPR 900 respectively.

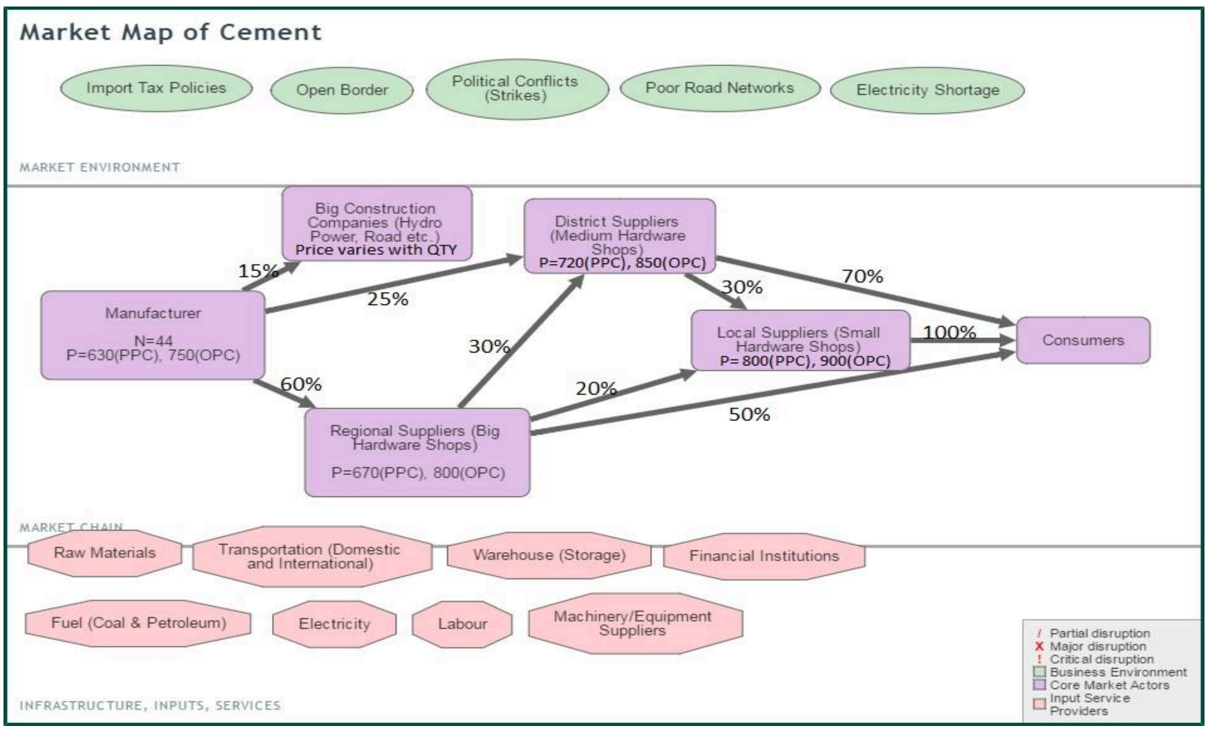


Figure 205: Market Map of Cement

TMT Bars:
Like the cement industry, manufacturers of TMT bars are located in Bhairahawa and Simra corridor of Nepal. The supply of goods is also exactly similar to that of cement industry i.e. 60%, 25% and 15% of their production is supplied to regional, district and big construction companies respectively. The ex-factory price of TMT bars is

NPR 62.5 per kg. The only difference here is that few manufacturers like Panchakanya sell their goods through the depo system. In such cases goods are transported to depos which are present at prominent location across the country and from these depos goods are directly sold to consumers. These depos are owned by the manufacturing company themselves. The average price for one kg of TMT bars at depos is between NPR 68-70.

From the regional level 50% of the goods are directly sold to consumers and the remaining 30% and 20% are sold to district and local suppliers respectively. Considering the transportation cost and margin of these regional suppliers, one KG of TMT bars is priced at NPR 68 at the regional level.

At the district level, majority of the goods (70%) are sold directly to consumers and the remaining 30% are supplied to local suppliers at the VDC level. The price at this level for one KG of TMT bar ranges between NPR 70-75.

At the VDC level i.e. local suppliers sell all the goods directly to consumers. But the price at this level is very high as the cost of transporting these goods to various VDCs is very high. The price of TMT bar at this level ranges from NPR 85 -90 per KG.

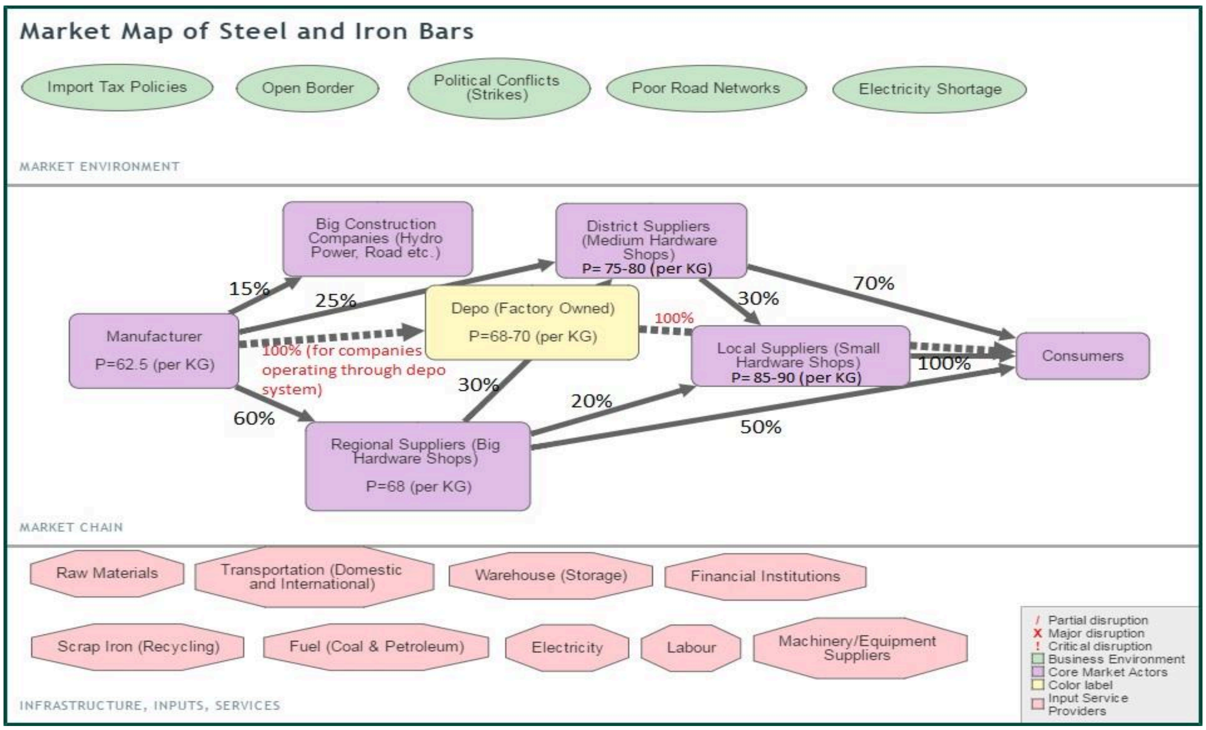


Figure 216: Market Map of TMT Bars

CGI Sheet:

Manufacturers of CGI sheets are located in Simra and Biratnagar corridors of Nepal. There are also imports of CGI sheets from neighbouring countries of India and China. From the manufacture level, 55% of the goods are supplied to regional suppliers, 35% to district suppliers and 10% are sold directly to big construction companies. Hulas like Panchakanya also sell their goods through the depo system. The price for 26 gauge heavy (approx. 62 kg) is NPR 7720 at the depo level.

Majority of the goods (60%) at the regional level is sold directly to consumers. The remaining 25% and 15% is supplied to local and district suppliers respectively. The price at the regional level for 26 gauge heavy plain CGI sheet is approximately NPR 7400 (Hulas).

From the district suppliers, 80% of the goods are sold directly to consumers and the remaining 20% are supplied to local suppliers. The price for 26 gauge heavy plain CGI sheet at this level is between NPR 7500 – 7700.

At the local supplier's level, 100% of the goods are sold directly to consumers. Price at this level for 26 gauge heavy plain CGI sheet (Hulas) is NPR 7800.

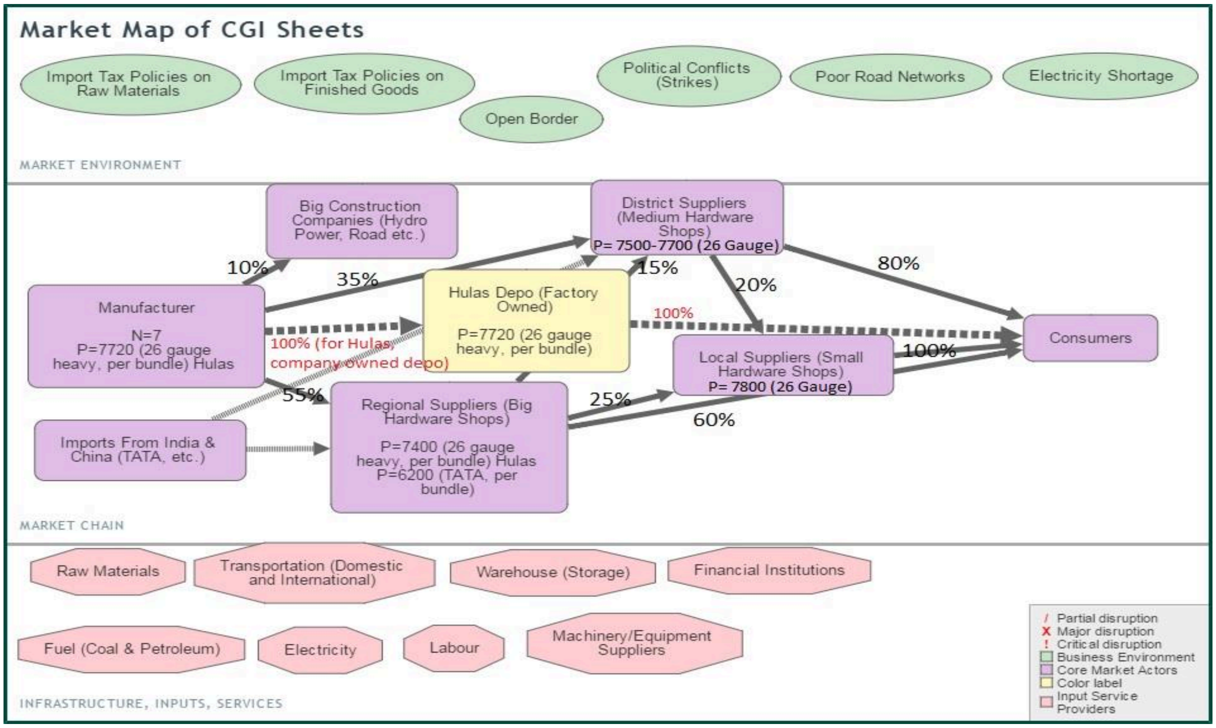


Figure 227: Market Map of CGI Sheets

Bricks:

In the brick industry there are very few depos and majority of the bricks are sold directly to consumers. Since there are usually three grades of bricks produced, price varies in regards to quality. At the manufacturing level (Chitwan District), price for grade A brick is NPR 14, grade B is NPR 10 and grade C is NPR 7. 80% of the production is sold directly to consumers and the remaining 20% is supplied to depos. These depos then add up the transportation cost and roughly about 10-15% of commission and sell them further to final consumers. The price for grade A brick at Bidur (Nuwakot) is NPR 18 per brick.

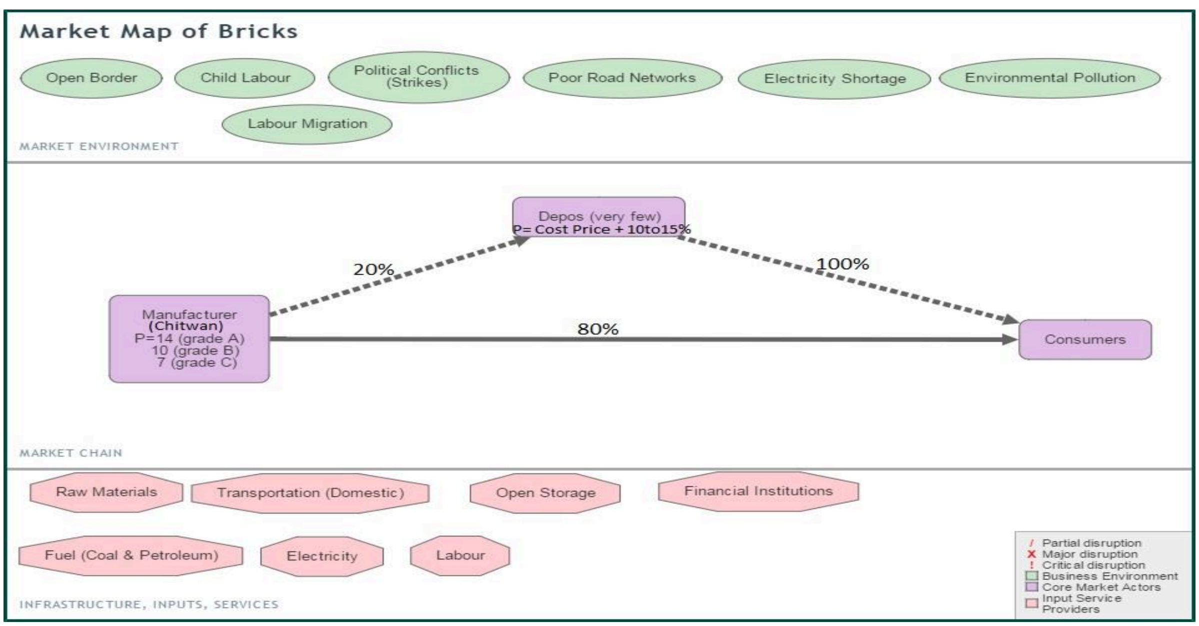


Figure 238: Market Map of Bricks

3.4.2.3 Local vendor's capacity

The are 45 hardware shops in Nuwakot and 5 vendors in Rasuwa district. Of the 45 hardware shops in Nuwakot, 37 are at Bidur, the district headquarter of Nuwakot. Rasuwa district have only 5 small scale vendors. The follwong table summarizes the combined monthly sale and storage capacity of the vendors in the study districts.

Table 13: Monthly sales and storage capacity of local vendors

Material	Unit	Sale Before the earthquake (Monthly)	Total Storage Capacity	Total in Storage
Rasuwa				
Cement	Bag	820	1800	157
CGI	Bundle	74	500	240
Rod	Kg	1200	15000	7500
Nuwakot				
Cement	Bag	27720	70230	22505
Rod	Kg	192790	552710	223830
CGI	Bundle	5057	19260	5195

The monthly sale record is of before the earthquake, so represents the sale in a normal time. The vendors reported that there was sharp rise in the sale of the construction materials, mainly the CGI sheet, immediately after the earthquake. However, it returned back to normal after few months. The sale even dipped during the blockade. Though the vendors believe that the demand for the construction materials will increase sharply once the reconstruction starts, they haven't increased their storage capacity. The vendors have the storage capacity of roughly 2.5, 3 and 3.5 times the monthly sale of cement, Rod and CGI sheet respectively. If it is considered as the monthly sale capacity of vendors and all the materials are supplied through the vendors, about 4 year will be required to meet the reconstruction demand. Likewise, about 2 years will be required to meet the demand for steel and CGI sheet. However, large number of households in the study districts purchases the materials from the regional and national markets. Bhairawa, Birgunj, Chitwan, Kathmnadu, Hetauda and Biratnagar are the major regional /national market for the construction materials for the district. Only a small number of materials come from kathamdu, as Kathmandu also imports the materials from regional markets and transpiration cost from the regional market to Nuwakot is cheaper than that of from Kathmandu. Likewise, small quantity of CGI sheet, about 6.6% also come from India. The following table summarizes the weightage of different markets, in terms of volume of materials imported to the study districts.

Table 14: Percentage share of different markets in terms of supplying materials to Nuwakot and Rasuwa

District	Materials	Bhairawa	Birgunj	Butwal	Chitwan	Kathmandu	Hetauda	Simra	India	Biratnagar
Nuwakot	Cement	31.3	47.1	7.8	11.2	0.4	2.2			
	CGI		40.2		9.6	0.9	4.4	38.4	6.6	
	Rod	23.1	54.4		22.1	0.3				
Rasuwa	Cement	33.3	44.4	11.1						11.1
	CGI	41.7	47.2	5.6						5.6
	Rod	33.3	44.4	11.1						11.1

Figure 29: Nuwakot Vendor Map

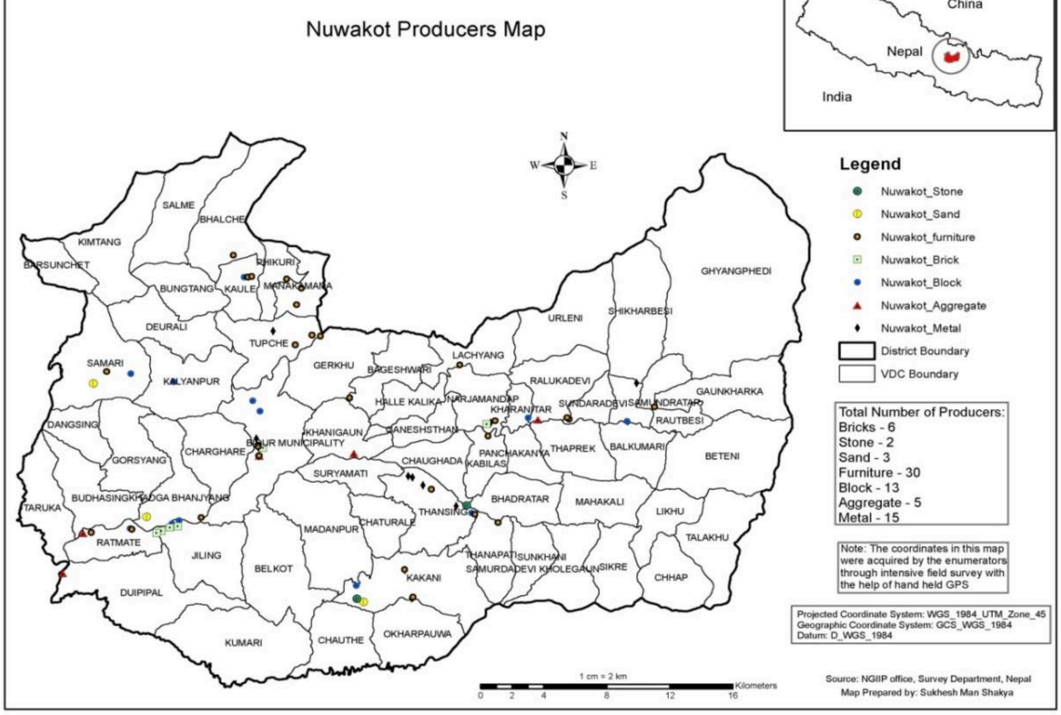
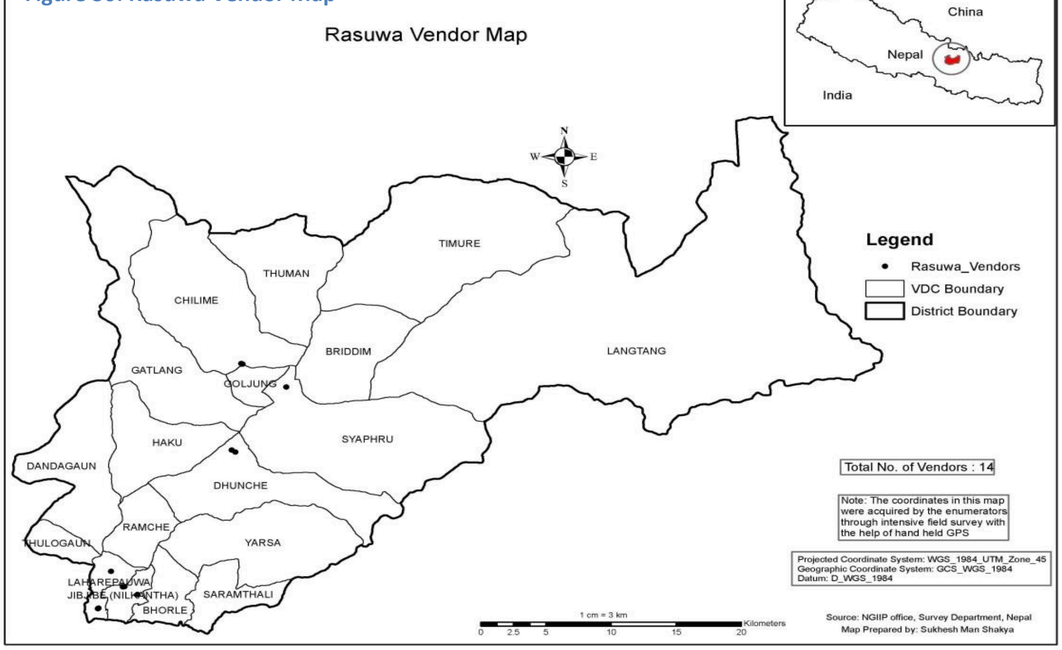


Figure 30: Rasuwa Vendor Map



3.4.3 Factors affecting price of construction materials

It is anticipated that the price of construction materials will surge up during reconstruction phase due to mismatch of demand and supply. The rise in the price can be attributed to both temporal and spatial dimensions. The rural areas with poor road infrastructure facility will have a higher chance to see substantial rise in price of construction materials. The reason could be attributed to the surge in cost of transportation as well as surge in price of construction materials. The season has also direct bearing on the cost of transportation in rural areas. During rainy season the charge of transportation will be higher by 100 Rs/Km/ per triper compare to that of other season. So, cost of materials depends on seasonal variable as transportation cost varies accordingly.

The price immediately after earthquake hiked for most of the construction materials particularly price of CGI sheet due to high demand from development organization and government to distribute as immediate relief. A comparision betwen current price of construction materials and price just before earthquake indicates that price level is higher by 4- 13 % in different construction materials. The most significant rise is in Cement.

Table 15: Average prices of materials before and after earthquake of non-local materials

Nuwakot	**Unit**	**Before**	**After**	**Percentage Change**
Cement	Per Sack	737	836	13.4
CGI Sheet	Per Bundle	5800	6200	6.9
Rod	Per kg	73	78	7.3
Rasuwa				
Cement	Per Sack	817	883	8.2
CGI Sheet	Per Bundle	6200	6650	7.3
Rod	Per kg	81	84	3.7

The section 3.4.2 has estimated the expected demand of construction materials and current supply capacity. Based on this, it is expected that gap between demand and supply will widen further during reconstruction. This will definitely affect the level of price in the market. Followings are some key issues in determining price level of construction materials.

Expected price of non-local construction materials: The current production capacity of CGI sheet is more than enough to meet the national demand during reconstruction. However, the past and current trend in CGI sheet indicates that around 50% domestic production is exported to India. If this is not discouraged for a time being, the price of CGI sheet may increase. Simiarly, the Cement and Iron rod factories are operating under 50% of its total capacity due to shortage of power. If the situation perpetuates in the reconstruction phase only way to meet the demand of cement and iron rod is to import from India. Import of construction materials from India is a common practice even during normal time. Thus, it is expected that price won't go up that much.
However, due to poor distribution system within country particularly in remote areas, shortage of construction materials may occur at a point of time and at a specific place evenafter ensuring national level supply to meet the national demand during reconstruction. This might create price surge in specific location which affect poorest of the poor living in remote areas.

Expected price of local materials: The expected demand and supply situation in the section 3.4.2 indicates that there will be a short supply of local construction materials. Especically, the supply of wood, sand and aggregates will be in short supply. Whether a particular district is in position to supply such local materials depends on availability of natural resource, number of local enterprise, their existing capacity and access to finance to expand their business. The short supply of local construction materials will be more serious than non-local construction materials as many existing government law and provisions prohibit entrepreneurs to expand their business by harnessing natural resources. In this context, the price hike in local materials will be higher compare to non-local materials. It's not only the price but more importantly the quality of materials may diminish during reconstruction as local raw materials will be in short supply and local entrepreneurs may use inferior raw materials. Based on survey team consultation with concerned stakeholders in the districts the amount of deficit on local materials will reach around 50%. The price effect of such shortage is expected to be around 30% hike in price of construction materials.

3.5 Key Issues in Supply Chain Management

The study indicates that the current level of supply capacity of the non-local and local construction materials is insufficient to meet the demand of materials during reconstruction phase. Besides, ensuring quality of the construction materials at the competitive price is a huge challenge. The study have analyzed some the major issues and constraints, which warrant urgent attention and actions, to improve the supply chain situation in order to meet the reconstruction demand. The issues are discussed below,

3.5.1 Issues related to manufacturer level

The following table summarizes the issues on the production of the major construction materials at the manufactures level.

Table 156: Issues related to manufacturer level

Materials	Major issues
Cement	**Power shortage**: Due to insufficient power supply (load shading), the factories are running only at 60% of their capacity. **Raw materials –** High dependency on India for the raw materials. About 30% clinker Is imported from India. Trade relation between Nepal and India has direct bearing on access to raw materials. **Certification:** Certification provision-1995 doesn't' allow certifying higher grade cement in the country. Only 33- grade cement can be produced in the country, though, the factories have capacity to produce higher grade cement. As a result, the major development project, hydropower, imports the higher grade cement from outside the country. This dissuades the local factories to increase their capacity
CGI sheet	**Power shortage**: Power shortage is one of the major problems in CGI sheet as well. Factories have started to install high capacity generator to meet the power requirement. However, it will increase the cost of the production and the price of CGI sheet **Export to India**: Around 50% CGI produced in Nepal is exported to India, which might result a shortage of CGI sheet during reconstruction phase , though the local production is more than sufficient to meet the local demand. The government had put ban on the export of CGI to India for some time after the earthquake. However, the ban has been lifted now. **Government policy**: The government has relaxed the import tax by 15% on the import of CGI sheets. However, it hasn't relaxed the taxes on the raw materials.
Bricks	**Skilled labor**: The shortage of the skilled labor is the main problem in the production of brick. Majority of the skilled labor come from India. The labor has to be booked in advance. Likewise, the child labor is another problem with Brick production. Studies suggest that about 34% of the labors in Brick Kilns are underage (Health Research and Social Development Forum, 2016) **Quality**: Majority of the Bricks produced in the country doesn't need to meet the quality standard. Compressive strength of the good quality brick should be at least 5 N/mm2 however majority of the brick produced in the country doesn't meet that requirement (below 3 N/mm2). **Technology:** Majority of the bricks are produced traditionally, in a Fixed Chimney Bull Trench Kilns (FCBTK), which consumes 4 time higher energy than Vertical Shaft Brick Kiln (VSBK), the modern technology in brick production. Production capacity of the traditional Kiln is also low. **Unhealthy competition:** Unhealthy completion among the brick Kilns is another factors that is compromising the quality of the production. In a bid to provide cheaper to reduce the prices, the Kilns are compromising with the quality of bricks.
Steel rod	**Shortage of electricity**: The steel production is also plagued by the shortage of electricity. The steel mills are running at about 66.5% of their potential capacity (*NRB Study, 2014*) due the shortage

	of power. **Trucks and loading/unloading equipment's**: The storage of larger trucks for transportation of steel bars and loading and unloading equipment is also a constraint in the supply of steel bars.

3.5.2 Issues related to transportation

Majority of the supply chain actors has identified poor transportation service as one of the major constraints in the supply chain management. The overwhelming, 86% of the people in Nuwakot and 100% people in Rasuwa have identified the transportation as one of the major difficulties. There is only 9.2% and 8.6% of road blacktopped in Nuwakot and Rasuwa district respectively. The gravel and the earthen road often get disrupted during rainy season. Hence, the bad road condition is the major problem. Such situation not only hampers the timely delivery of the construction materials but also increases the transportation cost with high transportation fee. During the monsoon, per km transportation fare can raise up to Rs 125/km. Likewise, the transportation charge is haphazard and high which significantly raises cost of the materials transportation. The cost of transportation varies Rs 200 to 450 /km in Nuwakot and Rs 300 to Rs 550 in Rasuwa (see table 17).

The survey with transporters indicated that all transporters are not the member of transport union. Around 46.2 % transporters are not member of transport union. Hence, it looks like that the transportation union has not been able to accommodate all the transporters. Some transporters raised concern on administrative hassle during transportation. Likewise, due to the bad road condition, the breakdown of the vehicle is another issue which is affecting the transportation. As there are no good maintenance facilities in the rural areas, the fixing of damaged vehicles takes longer time. Some of the transporters also complain about the problem of parking. The transportation loss is around 2 % and there is no practice of insuring goods during transportation.

Table17: Cost of Transportation

VDC	From	To	Road Condition	Distance	Time (In Hrs)	Type of Vehicle	Rate per KM
Nuwakot							
Deurali	Trishuli	Deurali	Rough Road	22	4	Mini Truck	450
	Kimtang	Deurali	Rough Road	25	5	Bus	450
	Barsunchet	Deurali	Rough Road	27	5	Mini Truck	500
Kaule	Trishuli	Kaule	Rough Road	20	1.5	Mini Truck	400
	Bidur NP	Bhalche	Rough Road	27	2	Mini Truck	370
	Bidur NP	Manakamana	Rough Road	18	1.15	Mini Truck	350
Narjamandap	Narjamandap	Bhalche	Rough Road	15	1	Mini Truck	400
	Narjamandap	Kathmandu	Black Top	45	4	Mini Truck	400
	Narjamandap	Tupche	Black Top	8	1	Mini Truck	400
Balkumari	Tupche	Bidur NP	Black Top	35	3	Mini Truck	200
	Khanigau	Rautbesi	Gravel	35	4	Mini Truck	300
	Samundratar	Shikharbesi	Gravel	30	3	Mini Truck	300
	Samundratar	Gaunkharka	Gravel	50	5	Mini Truck	350
	Tupche	Kathmandu	Black Top	75	5	Mini Truck	200
	Samundratar	Samundratar	Black Top	18	1	Mini Truck	200
Thansing	Thansing	Kathmandu	Black Top	35	4	Mini Truck	450
	Thansing	Bidur NP	Rough Road	14	1.5	Mini Truck	450
	Tansing	Bhadrutar	Rough Road	10	1	Mini Truck	500
	Suryamati	Tupche	Rough Road	30	2	Mini Truck	350
Okharpauwa	Kathmandu	Bidur NP	Black Top	75	4	Mini Truck	300
Bidur Municipality	Trishuli	Jiling	Black Top	10	1.5	Mini Truck	300
	Duipipal	Bidur NP	Black Top	25	1.5	Mini Truck	200
	Khanigau	Bidur NP	Black Top	12	1	Mini Truck	200

	Bidur NP	Khadag Bhanjyang	Black Top	10	1.5	Mini Truck	300
Budhasing	Trishuli	Budhasing	Rough Road	25	2	Mini Truck	200
	Bidur NP	Gorsyang	Rough Road	30	3	Mini Truck	200
Rasuwa							
Laharepouwa	Kathmandu	Syafrubesi	Black Top	162	6	Mini Truck	500
Chilime	Kathmandu	CHILIME	Black Top	150	3	Tripper	550
Dhaibung	Kathmandu	DHAIBUNG	Gravel	185	8	Tripper	500
Bidur	Trishuli	Dhunche	Rough Road	45	3	Tripper	400
Municipality	Trishuli	Saramthali	Rough Road	30	3	Tripper	400
Laharepouwa	Kamidanda	DHAIBUNG	Black Top	15	2	Mini Truck	300

The study further conducted KII with transporters to have a better understanding of the issues. The following table summarizes the issues in transportation from the transporters' perspective.

Table 18: Priortisation of issues related to transportation

Difficulties	Percentage
Road condition	84.6
Unnecessary administrative interference	61.5
Absence of transportation groups	46.2
Parking Problem	15.4
Maintenance Problem	15.4

3.5.3 Issues related to quality

Quality of the construction material is one of the major issues. The 55% of people in Nuwakot and 69% people in Rasuwa complains that the construction materials are sub- standard. The vendors also agree that there is influx of sub-standard construction materials after the earthquake. It is most evident in CGI sheets. However, the risk of the sub-standard materials is higher in the local materials. Ironically, it is little appreciated by the households and the authorities. The non-local materials (Cement, Rod, CGI sheets) are branded and have to be NS/ISO certified which limits the adulteration of the non-local materials. However, the local materials are commonly managed by the house owners themselves. Majority of the house owners are not aware about the minimum standard of the local construction material (Sand, aggregates), though Nepal Building Code (NBC) have the prescription about the materials. Even the local enterprises of the materials are not much aware about the quality aspects or are negligent about it. The majority of the aggregate crushers in Nuwakot are selling the pebbles from the rivers as aggregates. The pebbles are not only rounded and polished, which makes them unfit to be used as aggregate, the mix is also not right (majority of the pebbles are large). Likewise, silica content is very high in the sand that the sand mines are selling. On the other hand, majority of the people are still planning to use random rubbles in the wall construction. The random rubbles are one of the major reasons which make the traditional houses weak. Hence, there is urgent need to educate people about the risk of using random masonry in an in proper way and equip them with the skills and technology to make the rubbles regular. Similarly, as discussed in the previous section, the salvaged materials are the major source of materials during the reconstruction. However, all the salvaged materials, particularly wood, will not be suitable to use in new construction. The risk is higher considering the fact that about 1.5 years have already elapsed after the earthquake (hence some of the wood might have already rotten). Besides, there is risk that some people will use the unseasoned wood during the reconstruction which will severely compromise the safety and strength of the building.

Besides, the quality of the brick and blocks, is concerning. Majority of the bricks produced in the country have a lower compressive strength then 5 $N/mm^{2,\ \text{which}}$ a good quality brick requires. Likewise, the concrete block industries that are mushrooming in Nuwakot district after the earthquake are not adhering to any standard. Each

block industry uses their own thumb rule for mixing ratio (Cement, stone chips/pebbles, Sand). One thing is common in all the block enterprise, which is the low use of cement to reduce the cost of production.
No institutions are monitoring the quality of the construction materials at the moment at district level. The market monitoring committee at District Administration Office is more concerned about the monitoring of the food items and medicine. Though the committee is ready to take on the responsibility for the construction materials as well , it lack the required skill and the equipment.

3.5.4 Legal Issues

There are some legal issues which will directly affect the supply system of construction materials. The assessment has identified such issues and discussed below.

For local materials:

***Forest act 1993 & Forest Regulation 1995*:** The act and regulations prohibits cutting trees, extracting sand and aggregates from National, protected, leasehold and community forest other than prescribed in the work plan. The department of forest prepares and ministry of forest approves such work plan. In case of community forestry, the forest users groups prepare the plan and the district Forest officer approve it. Preparing the work plan and getting approval is a cumbersome process.

***National Parks Wildlife Conservation Act, 1973 (Section 5)*:** Prohibits the followings without written permission from the authorized official,

- To cut, clear, fell, remove or block trees, plants, bushes or any other forest resources, or do anything to cause any forest resources dry, or set it on fire, or otherwise harm or damage it,
- To dig mines, stones or remove any mineral, stone, boulder, earth or any other similar

***Mines and Minerals Regulation act 1957*:** Section 19, Sub section C prohibits to perform excavation in the places allocated for the national and public interest and safety and within at least Fifty meters from an ancient monument, city, grave yard, crematorium, public way, dam, canal, pipeline, fort, army barrack, temple, mosque, church, house, factory etc. Section 11 A, 32 E put the condition that the sand mines/operators shouldn't use heavy equipment that my cause vibration and noise pollution and not to construct a house, factory an area.
Likewise, in July 2014, the government through a cabinet level decision established a new standard for crusher industry which requires the crusher industries to be placed at least 500 meters away from highways, 500 meters away from riverside, at a distance of 100 meters from high-tension electricity grid and 2 km away from schools and colleges, health centers and hospitals, security installations, forests, national parks, human settlement and religious and archaeological site (*The Kathmandu Post, July 2015*) . The standard hasn't come fully inforce due to protest from Crushers owners. However, if the standard is enforced, very few crushers industries will survive. Besides, the district authorities has imposed a narrow window period of 6 months for sand extraction

***Environmental Protection Regulation- 1998*:** The regulation requires the Initial environmental examination (IEE) and Environmental Impact Assessment (EIA) for following enterprises.

IEE

- Establishment of saw-mills which could to process 5000 to 50,000 cubic feet of timber per year.
- (Collection of boulders, gravel and sand and extraction of coal and other minerals from forest areas.
- Establishment of boulder crushing industries.
- Establishment of bricks and tiles industries with a production capacity of 10 million units per year.
- Excavation of construction oriented minerals in small scale,
- Extraction of 10 to 50 cubic meters of sand, gravel and soil from river beds per day.

EIA

- Establishment of saw-mills processing more than Fifty cu ft. of timber per year.
- Establishment of saw-mills, bricks and tiles factories and tobacco processing industries within Five Km. from the forest boundaries.
- Establishment of brick and tiles industries with a production capacity of more than Ten million pieces per year.
- Extraction of sand, gravel and soil at the rate of more than fifty cubic meters per day from the surface of river and revolute.

Non local Materials:

***Certification provision-1995*:** The certification provision 1995 prohibits the higher grades cements in Nepal. Only 33 grade cement can be certified in Nepal. Hence, the major development projects including higher grade cement (43, 53) import cement from India. Though the 33 grade cement is good enough for individual houses and majority of the houses will use them, the provision discourages the cement factories to increase their overall capacity.

***Relaxation on the import tax on CGI sheet*:** The government of Nepal has relaxed the import tax on CGI sheet by 50 % (Before it was 30% of the value, now reduced to 15%) following the earthquake. However, it hasn't relaxed the tax on raw materials, which is discouraging to the local manufactures.
Policy on import of CGI Sheet: The current production of CGI is sufficient to meet the regular and the reconstruction demand. However, about 50% of the production is exported to India, which results deficit of CGI sheet in the country. After the earthquake, the government of Nepal imposed ban on the export of CGI sheet for some time. However, the ban has already been lifted.

3.5.5 Issues related to aggregation

Due to the transportation difficulty, the transaction cost is very high. Aggregation of demand can reduce the transaction cost. Majority of the people in Nuwakot and Rasuwa agrees with the idea. However, they see few challenges to execute the idea. The following table summarizes the challenges.

Table 19: **Challenges in demand aggregation**

Challenges	Nuwakot	Rasuwa
Lack of storage space	43.6	62.5
Difficulty to work in group	47.3	62.5
Trust Issues	43.6	43.8
Difficulty in collecting money	49.1	68.8
Internal conflict	38.2	56.3
Scattered settlements	40.0	68.8

Households identified the lack of storage space as of the challenges. As majority of the public facilities, which are traditionally used for the storage are damaged by the earthquake, there are no space for the storage of large quantity of materials. Hence, support for the storage is needed.

Likewise, the aggregation requires the households in a community to follow a same schedule for house construction. Hence, the communities will require support for planning and scheduling of the house construction. Scattered nature of the settlement is another issue, which will require the aggregation to take place at the smaller units . Mechanism could be developed to further aggregate the demand from smaller units to VDC or Cluster level.

3.5.6 Issues related to storage capacity

The material wise total storage capacity of the local vendors is presented in the table below. The quantity of the materials in the store at the time of the survey has also been recorded.

Table20: Average storage capacity and current quantity in store of non-local materials

Material	Unit	Total Storage Capacity	Total in Storage
Rasuwa			
Cement	Bag	1800	157
CGI	Bundle	500	240
Rod	Kg	15000	7500
Nuwakot			
Cement	Bag	70230	22505
Rod	Kg	552710	223830
CGI	Bundle	19260	5195

The vendors have the storage capacity of roughly 2.5, 3 and 3.5 times the current monthly sale of cement, Rod and CGI sheet respectively. However, once the reconstruction pick the speed, the sale of the construction materials, will raise significantly. Hence, there the deficit of the storage space at the vendor level is eminent. Besides, as discussed in section above, there is storage problem at the community level.

3.5.7 Issues related to capital/finance

People are willing to build the earthquake resilient building. However, the cost of the government approved earthquake resilient building is much higher than the traditional house. Majority of people in the district think that the government direct financial support (NRs 200,000) wouldn't be enough even to cover half of the cost. Very few people have saving or asset to cover the extra cost. Likewise, relatively small number of people earns enough remittance to support the house construction. Hence, the large number of people, 48.3% in Nuwakot and 58.4 % in Rasuwa are planning to take loan from banks to cover the additional cost. Majority of the people are eyeing for the soft loan that the government has provisioned for the earthquake victim

Table21: Fund Management

Sub-Cluster	Take loan from friends/relatives	Sell fixed assets	Use own saving	Utilize remittance	Loan from bank	Other
Nuwakot						
Urban Type	12.0	21.5	5.2	17.0	44.2	0.0
Semi-Urban Type	14.5	5.2	10.3	14.6	53.9	1.5
Rural Type	30.7	5.0	9.5	8.2	40.4	6.3
Overall	**18.0**	**8.9**	**8.9**	**13.5**	**48.3**	**2.4**
Rasuwa						
Urban Type	7.7	5.8	2.5	10.0	71.5	2.5
Semi-Urban Type	17.9	2.9	4.9	10.2	55.9	8.1
Rural Type	29.5	7.6	2.9	6.3	49.7	3.9

Overall	19.2	5.7	3.4	8.7	58.4	4.8

The study also explored the issues related to access of capital among local vendors to expand their business. Most of the village level vendors and entrepreneurs indicated that accessing capital for investment in the business is the most important issue in the business. However, capital is not a major issue for vendors at the district level.

3.5.8 Issues related to technology

The local level producer and entrepreneurs have limited access to the modern technologies due to poor knowledge and higher cost of technologies. Hence, the majority of the sand mines, aggregate crushers and the brick Kilns are owned by outsiders. The lack of knowledge on modern technology has inhibited the local producers, mason and the people using the mordent tools and equipment, which can improve the quality of the construction materials. For instances, people doesn't know about the wood treatment technology.

4. Recommendation

A. Recommendation on increasing access to construction materials

Materials	Major constraints	Recommendation	Responsible Agency
Stone	Government act /regulations like Forest act 1993, National Parks and Wildlife Conservation Act, 1973, and Environmental protection regulation, impose restrictions on stone quarries	Review/revisit the acts /regulations, relax the provisions at least for reconstruction period in a way it doesn't have significant impact on the environment, also encourage measures for replenishments	Ministry of Forest , Ministry of Industry , NRA
Sand	Narrow window period for sand extraction (3 months)	Increase the sand extraction window at least for reconstruction period	DDC
	Restriction on the use of heavy equipment	Relax the provision at least for reconstruction period	DDC ,Mines
	Mines and Minerals Regulation act 1957 requires the sand mines to be at 50 m away from public facilities, which is impractical, considering the fact that the roads often run along the river bank	Review the law and make it more friendly to meet the reconstruction demand with adequate environmental protection provision	Department of Mines, Geology
	High dependency on the river/stream (91% in Nuwakot, 82% in Rasuwa)	Identify the safe sand mines sites (use experts) and expedite the mine registration process, educate the local communities about safe mining practice	Department of Mines, Geology, DDC, Expert organization, Private sector
Timber	Forest act and regulation impose a lot of restriction on Timber harvesting, complete prohibition in national park	Relax/suspend some the clauses at least for the reconstruction period with minimum consequences on environment degradation	Ministry of Forest and Soil Conservation, Department of forest, NRA, National parks
	Locally available woods are	Promote RCC slabs for bands in urban and	POs

	inferior in quality to be used for buildings	Semi Urban areas	
		Promote Bamboos - for roofing (truss), partition and floor in rural areas	POs
Cement	Power shortage - load shading	NEA to provide uninterrupted power supply to the factories at least for reconstruction period or tax or import duty relaxation on import of high power generators	NEA, Ministry of Finance/ Department of Revenue, NRA
	Lack of certification provision for higher grade (43, 53)	Amendment of certification provision-1995	Nepal Bureau of Standards and Metrology.
	High dependency on India for Raw materials (clinkers)	Encourage and support the cement factory to have their own Clinker plant	
Reinforcement Rod	Power shortage - load shading	NEA to provide uninterrupted power supply to the factories at least for reconstruction period or tax or import duty relaxation on import of high power generators	NEA, Ministry of Finance/ Department of Revenue, NRA
CGI Sheet	Power Shortage - load Shading	NEA to provide uninterrupted power supply to the factories at least for reconstruction period or tax or import duty relaxation on import of high power generators	NEA, Ministry of Finance/ Department of Revenue, NRA
	Export to India - 50% of the production	Ban on export of CGI during reconstruction period	
	Import of raw materials	Subsidy in import of raw materials	
Brick	Shortage of skilled labors-dependent on Indian labors Damage of brick Klins – 70% need renovation – NFBN	Develop the local skills	Cottage and Small Industries Development **Board** (CSIDB), POs

B. Recommendation on issues related to quality of construction materials

Materials	Major constraints	Recommendation	Responsible Agency
Stone	Use of random Masonry in the construction	Awareness about the importance of regular masonry, corner stone, through stone, support technology for preparing regular shaped stones	CSIDB, POs
	Corner and through stone	Promote stone cutting enterprise at the local level	CSIDB, POs
Sand	Sand extracted from river have high silica content	Develop ICT materials about the standard and quality of sand, provide training to producers, awareness raising program for consumers, periodic monitoring	POs for ICT and awareness, District Administration Office and DUDBC for monitoring
	Use of larger grain sand	Develop thumb rules for quality testing (*like of doctor Navayama*)	POs, NSET
Aggregate	Use (Round /polished)pebbles as aggregate	Encourage crushed aggregate , prepare and disseminate ICT materials on the quality of aggregation and quality monitoring encourage crushed aggregate, quality monitoring	NSET/Pos for ICT and awareness, District Administration Office and DUDBC for monitoring
	Larger sized aggregation and improper mix of different sized aggregate	Train the producers, prepare and disseminate ICT materials on the quality of aggregation and quality monitoring	

Wood	Use of un-seasoned and untreated woods	Make the chemical (Boric acid) for treatment locally available, promote other alternative cheaper alternatives like polishing it burnt vehicle Engine oil and traditional method like smoking the timber	NTB, FNCCI, POs
Brick /Blocks	Strength of brick not up to the standard- a good quality brick requires to have at least 5 N/mm2 compressive strength , however, the majority of brick available in the market only have 3 N/mm^2	Develop the strength testing facilities at the local level, (ideally with DUDBC and DTO at the district level) test the strength on periodic basis and at IoE a the central level	DUDBC, POs, IOE
		Provide training to the operators and labor to improve the quality	CSIDB, POs
		Stringent action against the low quality brick producers	District Administration office
		Awareness programme and simplified tool kit for quality testing for consumers	Pos/NEST
	No formal institution for the research on earthquake/construction materials	Identify NSET as the research institution	NRA, NSET , MOST
		Engage Academia (Like IoE) in research on construction materials	DUDBC, NRA,POs
		Develop information Centre at the Resource Centre of NRA for dissemination of construction materials related information	NRA, POs Academia
	Absence of Robust monitoring Mechanism	Active the market monitoring committee at the district level for monitoring of the construction materials	Ministry of Home, DAO, DUDBC

C. Recommendation on issues related to price of materials

Materials	Major constraints	Recommendation	Responsible Agency
Non local materials (Cement, Rod, CGI and Bricks)	Production cost high due to higher hours of load shading , use of generators, import of raw materials and labors (in case of Brick)	Continue supply of electricity, subsidy on generators and raw materials and develop local human resource	
	Hoarding of materials at factory and regional level	Periodic market monitoring	Ministry of home, CSIDB, Pos
	Very high transportation cost- a bag of PPC Cement which cost Rs 715 at Bidur, Nuwakot, cost Rs 890 at Kaule VDC. Transportation cost from Bidur to Kaule is Rs 100 per bag of cement	Aggregate demand and develop construction materials dealers at the local level	FNCCI, Chamber of Commerce, Pos
		ICT based price information dissemination system	FNCCI, Ps
		Differential subsidy for Rural, Urban and semi urban areas or additional subsidy for rural areas for transportation	NRA
Local materials	Haphazard pricing	Develop scientific basis for pricing of the local materials and enforce	FNCCI, District Administration office, DDC
	Periodic interruptions on production from local authorities	Non –interference	DDC ,District Administration Office

D. Recommendation on the issues of transportation

Materials	Major constraints	Recommendation	Responsible Agency
All kinds of construction materials	Frequent disruption of transportation due to bad road	Give priority to the maintenance of road in the earthquake affected districts	DOR, DOLIDAR,

	condition		DDC
	Haphazard and exorbitant fare	Develop scientific basis for determination of transportation fare , enforce and monitor, subsidy on transportation fares at remote areas (if possible) - BT -286, Gravel-317, Rough -422	DDC, FNCCI, DAO , NRA
	Absence of vehicle maintenance facilities at the rural areas	Develop maintenance facilities at remote area as well (*can be promoted as local skill based enterprise*)	FNCCI ,

E. Recommendation on issues related to storage capacity

Storage	As most of the public facilities, VDC building, School or community building, which are commonly used for storage during emergency, have been damaged, there is limited storage capacity at the local level	Support to develop temporary storage facilities or develop collection facilities which are used for other public purpose after the reconstruction	DDC, Pos, DUDBC

F. Recommendation on establishment of construction material enterprises at the local level

Construction materials enterprise	Registration process, mainly that of the natural resource based is cumbersome	Make the process less cumbersome	CSIDB, DFO
	Electricity- The enterprises requires 3 phase electricity line. Connecting the 3 phase line is a lengthy process if it is done at NEA cost and expensive if it is done at entrepreneurs cost	Expedite and facilitate the process	NEA , NRA
	The local people have limited capital to establish the enterprises at local level	Provide subsidized loan to the construction material enterprise,	Banks
	Limited technical and entrepreneurship skills among the local entrepreneurs	Provide technical and entrepreneurship training and support to establish forward and backward linkage	Pos, CSIDB

G. Recommendation on issues related to local vendors

	Limited storage capacity	Support to increase the storage capacity	FNCCI, PO,
	Number of vendors and their capacity not enough to meet the reconstructions demand	Support to increase the capacity of the local vendors and support to develop new vendors at the local level	FNCCI, PO,

	The local vendors doesn't have enough capital to expand the business	Subsidy, subsidized loan and establish linkages with the financial institutions	NRA, FNCCI, PO, Financial institutions

Besides the above specific issues, there is also need for a recognized research institution at the national level, which leads and coordinates the researches on the earthquake. Likewise, a mechanism is also needed at the central level, comprising the representatives from concened government ministries and othe relavent stakeholders to monitor the supply chain of the construction materials at the national level.

5. Strategies for strengthening the supply chain of construction materials

The demand and supply gap, quality, price and timely supply are the four major issues related to the supply chain of construction materials. An effective supply chain management strategy should devise appropriate measures to address the issues. Some of the possible measures are discussed in this section.

5.1 Supply and demand gap

The study indicates a huge gap in supply and the demand of the construction materials. Hence, an effective strategy to reduce the gap would be to adopt the measures to reduce the demand and increase the supply of the materials. The followings are some of the measures,

5.1.1 Reduce the demand

Usage of salvage materials: The study has shown that about 50% of demand for stone and 30% of demand for wood can be met through salvaged materials from the old buildings. The house owners already regard the salvaged materials as one of the main source of materials. However, awareness programs on the strength and usability of the salvaged materials is required as all the salvaged materials are not strong enough and suitable for new construction.

Retrofitting: A study carried out by Build Change Nepal, estimates that about 25% of the damaged houses can be retrofitted. As the retrofitting requires very less amount of construction materials in comparison to new building, it can reduce the demand for materials significantly. On the other hand, it can save money. The same study maintains that the cost the retrofitting is almost 6 times cheaper. However, not many people are aware about the possibility of retrofitting and retrofitting techniques. Hence, retrofitting drive should imbed awareness and encouragement programme besides the support for technology. Likewise, government should also come up with the appropriate subsidy mechanism to encourage retroffing.

Promotion of alternative materials: The source of traditional construction materials like stone and wood are limited and the conventional materials like brick, cement are beyond the means of many people. Hence, it is important to promote the alternative materials which use local raw materials and can be produced locally. Various organization/institutions have been promoting various kinds of alternative materials after the earthquake. However, they are promoting their products disparately. The government of Nepal will need a coherent policy for the promotion of the alternative construction materials. A study carried out by Practical Action, has shortlisted 5 alternative construction materials for scaling up. The materials have been selected based on the criteria like structural strength, affordability and adaptability.
The materials mostly use the local raw materials and can be produced locally using the local human resources. Hence, the materials can be promoted as the local enterprises and liked with the rural livelihood.

5.1.2 Increase the production

Sensible exploitation of natural resources: The natural resources are the only source of the local construction materials like stone, aggregate, sand and stone**.** Hence, people will rely on the resources for the construction materials during the reconstruction. However, indiscriminate exploitation of the natural resources may lead to unrepairable environmental hazards. Hence, it will be important to map the available resources and determine the maximum extraction limits. Once the sources are identified and the limits are determined, the government of Nepal should expedite the licensing process for the extraction of the materials. The forest and the mines & minerals laws as discussed in Chapter 3, is too cumbersome for a post disaster situation. In the meantime, the appropriate mitigation and replishment measures should be integrated in the material exploitation plan.

Non local materials: The assessment shows that Nepal is self-sufficient in the production of major non local construction materials like Cement, Rod and CGI sheet. If the existing factories of the materials operate to their full capacity, the reconstruction demand can be met with in a year. However, majority of the factories are producing at about 50% of the installed capacity due to power shortage. Hence, uninterrupted supply of the electricity will be important for increasing the production of non-local materials. Some of the factories have already imported the high capacity generators to meet the electricity demand. However, use of the generators, will increase the cost of production, which will ultimately lead to hike in price of the materials.

5.2 Price of the construction materials

The price of the construction materials can be kept competitive by reducing the cost of production, reducing the transaction cost and closely monitoring the market price. Each of the measures are briefly discussed below

5.2.1 Reduce the cost of production:

The irregular power supply is one of the main contributors to the high production cost. Hence, it is important to ensure the uninterrupted power supply to the factories to reduce the production cost. Likewise, the import tax waiver on raw materials will also help to reduce the cost of the production. Similarly, developing the necessary skilled human resources, will also help the in the production cost reduction (mainly in the brick production). Use of modern technology can also reduce the cost by increasing the production efficiency

5.2.2 Reduce the transaction cost

The followings are some of the measures for reducing the transaction cost

Demand Aggregation: The aggregation of demand or the group procurement can help to reduce the cost of the construction materials by reducing the transaction cost and increasing the bargaining power. If the demand aggregation can be done, there can be significant saving on the transportation cost which is the main contributors to the high price of the construction materials at the rural areas. Besides, the aggregated demand will provide larger choice of the markets and bigger bargaining power. If the quantity is larger, they can procure the materials directly from district, regional market or even from the factory at a discounted rate. The local level institutions like youth clubs, cooperatives and community level reconstruction committees which NRA is planning to set up can be mobilized at for the demand aggregation. The skilled masons (*Mistry*) and the local level vendors can also act as the demand aggregator. A preliminary assessment carried out by Practical Action in Nuwakot district shows that about 10-15% of the cost can be saved from demand aggregation. The following tables show a simple calculation for ward no 5 of Bidur municipality. It shows the difference in price when procured individually from the district and collectively direct from factory. The saving will be even more in the rural areas where the transportation cost is higher

S.No	Materials		QTY Required Per House (Brick with Cement Motar)	Discount Price (Aggregation)		Normal Price (Non Aggregation)	
	Name	Unit		From Factory	Total (Per House)	From District	Total (Per House)
1	Cement	Bags	**123**	707	86961	747	91881
2	Brick	PCS	**21253**	16.9	359175.7	19.6	416558.8
3	CGI	Bundle (50 KG)	**5**	5430	27150	5920	29600
4	Steel Bars	KG	**460**	64.5	29670	71	32660
TOTAL (Per House)					**502956.7**		**570699.8**

Dealership at the local level: Encouraging the dealership of the major construction materials at the local level can also help to reduce the price of the construction materials. The dealer's receive attractive commission from the manufacturers on meeting the sales targets. The commission encourages the dealers to sell the materials at bulk even offering the lower price (as the commission is more attractive than the higher margin). This will contribute to reduce the price of the construction materials.

Fixed the transportation fare: The following table shows the price of the construction materials at different level. Price of a bag of cement at Village level is almost 40% times more than the ex-factory price. The transportation fare mainly contributes to the price difference. As discussed in the previous chapter, the transportation fare is ad-hoc and varies significantly from place to place. It also varies with the seasons. Hence, the local government will need to fix a reasonable transportation fare and enforce it strictly. Besides, government should dismantle the syndication /carterling rampant in the transportation sector, to reduce the transportation cost.

Materials		PRICE					
		Factory	Regional	District	Depo	VDC (Semi Urban)	VDC (Rural)
Name	Unit	MRP	MRP	MRP	MRP	MRP	MRP
Cement	Bags	630	670	715		810	890
Brick	PCS	14	NA	NA	18	NA	NA
CGI	Bundle (50 KG)	5400	5650	5850	6090	6000	6200
Steel Bars	KG	62.5	68	70	73	75	80

5.2.3 Develop a monitoring mechanism

A close monitoring on the price of the construction materials will help to keep them at check. While government should take the prime responsibility for the monitoring of the price, it should be done at the different level

Market Information System (MIS): The user friendly market information system will provide the households access to the market information including the price. The access to the market information will enable the users to monitor the price of the construction materials themselves.

Set up a monitoring mechanism : It will be important to set up monitoring mechanism both at the district and central level. The mechanism should carry out the price monitoring on the periodic basis and take a stern action against the defaulters.

5.2.4 Mobilize the government institutions like national trading corporation /Salt trading corporation:

Strengthening and mobilizing the existing market system is the best strategy for the effective supply chain of the construction materials. However, the government/quasi- government institutions like National /Salt trading corporation can also be mobilized to supply the subsidized construction materials to the rural areas and price stabilization. The cooperation can act as the reference point for the price of the construction materials. At the meantime, the institutions can be instrumental to deliver the subsidized construction materials to the rural areas. The corporation can have their depots at the district level and distribute the materials to the rural areas through the local cooperatives. Government should determine the minimum quantity of non-local construction materials required for a rural housing and distribute the materials to rural households through the corporations and cooperatives

5.3 Timely availability of the construction materials

5.3.1 Improve the transport infrastructure: The bad road condition is one of the major hindrances for the timely supply of the construction materials. Hence, the government, mainly the DoLIDAR should give high priority in the maintenance of the roads in the earthquake affected districts. The maintenance of the road will not only help in the timely supply of the materials but also contribute in reducing the cost of the materials as the transportation fare varies with the road condition.

5.3.2 Strengthen the capacity of the local vendors: One of the strategies to ensure the timely supply of the construction materials bringing the vendors closer to the community. At present there exists limited number of vendors with limited capacity at the local level. Their capacity including the storage capacity is not enough to meet the reconstruction demand. Hence, efforts will be needed to develop more vendors at the local level and increase their capacity.

5.3.3 Availability of fuel: Frequent shortage of fuel may also hamper the timely supply of the construction materials. Hence, the government should provide preferential treatment to the earthquake affected district in the supply of the fuel.

5.3.4 Control hording of materials: The materials hording at the factory or vendors level to create the artificial shortage of the materials also hampers the timely supply of the materials. Hence, the government should strictly discourge the hording of the materials.

5.3.5 Transportation of the materials from one place to another place: Restrictions and tedious permit process, for transfer and transport of the local materials, mainly wood, will affect the timely supply of the materials. Hence, government should make the permit process more efficient

5.4 Quality of the materials:

Maintaining the quality of the materials is a big challenge when the demand for the materials is huge. The following measures are proposed for

5.4.1 Use of technology: Use of technology can improve the quality of the construction materials. Simple stone cutting technologies can help to produce the corner stones and through stones. Likewise, stone crushing technology are necessary for production of quality aggregate at the local level. Similarly, the wood treatment and seasoning technologies are necessary for the treatment and seasoning of wood.

5.4.2 Awareness about the quality of the construction materials: As discussed earlier in the report, people have limited knowledge on the quality of the construction materials. Numerous varieties and brands of construction materials are available in the market. People are at loss which brands are better for the house construction. Hence, it is necessary to aware people about different brand of materials. Enlisting of the best brands of the construction materials can also help the people to choose appropriate brand and varieties of the materials. Similarly, awareness is required on the quality of the local construction materials. Though there are standards on the quality of the local construction materials, the people have limited understanding about the standards. Likewise, the producers of the materials are not abiding by the compliance. Hence, it will be important to develop ICT materials on the quality and standard of the construction materials and dissimilate through various means. Likewise, the technical capacity of the local level producers

5.4.3 Monitoring Mechanism: An effective monitoring mechanism is essential both at the district and local level for the quality assurance. The market monitoring committee under the DAO at the district level can be mobilized for the quality assurance at the district level. Likewise, institutions with the representatives from the relevant stakeholders will be necessary at the center level as well. Likewise, a lab for testing the quality of the construction materials will be necessary both at the district and national level. A material testing lab should be established at DUDBC at the district level. At the national level, a well-equipped lab will be necessary at the Institute of Engineering. Besides, the issues wise strategies discussed above, a national level research institution dedicated for the research on the earthquake will be important for the overall coordination of the research and innovation on the earthquake including the construction materials.

ANNEX

Annex- 1: Survey questionnaire for earthquake victims

Practical Action

Supply Chain of Construction Materials and Tools in Earthquake Affected Districts- Survey 2016/2017

SURVEY WITH EARTHQUAKE VICTIMS

All information asked within this questionnaire will be used only for statistical purposes.

District: ____________________ Market Centre : ____________________

Stratum: ____________________ Interviewed Date:: ____________________

Name of Settlement: ____________________ Number of respondents: (Male...…..Female…....)

Interviewed by: ____________________

Note: Ask participants to map their community in a piece of paper and depict the following elments in the community map:

- Settlement location and road
- Natural resources (forest, river, lake, mines)
- Markets (Distance from settlement to market and availibility of transportation services)
- Cooperatives, Aama Association, Community Associations
- Source of local materialsAfter completion of the above mentioned task, discuss on the below mentioned topics and if possible include the information in the community map only.

Sir/Madam:

Dear respondent,

Practical Action is conducting survey on supply chain of construction materials and tools in earthquake affected districts- *Rasuwa* and *Nuwakot*. In this survey, primarily, situation of infrastructure damage at VDC level, supply and demand situation of different construction materials in the VDC, various issues related to reconstruction will be collected from key informant and will be assessed on the basis of the information obtained. This VDC is selected as a sample for this purpose. The information you provide are confidential by Statistics Act 2015 and they are only published collectively in which no personal records are visible.

I would like to request you to make this survey successful by providing correct and factual information to the enumerators who are coming for survey.

Thanking for your cooperation

Practical Action
South Asia Regional Office
Nepal

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A. INFORMATION REGARDING HOUSE DAMAGES AND RECONSTRUCTION

Q1. Provide detailed information regarding damages in your respective settlement. (Individual households and community buildings)

Q2. Description of houses (composition and structure) in the community before earthquake.

S. No	House Composition	Percent (%) of households having this composition in the community
1	Stone and Mud	
2	Stone and Cement	
3	Brick and Cement	
4	Others	

Q3. Awareness regarding the earthquake resistance model houses designed by the government.

Response Options	Frequency
Yes	
No	

(Check: total frequency = the total group members)

Q4. If you are aware, how many of you are planning to build the house as per government design? (%):

Q5. If not, what is the reason behind not building house as per government design?

Q6. When do you intend to start building your house?

SWA	BHA	ASHOJ	KARTIK	MANG	PUS	MAG	FAG	CHAI

Q7 Where are you planning to build your house?

Place	%
Same place	
Other place within same VDC	
Other (Specify)	

Q8. If getting compensation from government takes longer, then in that situation, what is your plan with respect to starting building your house?

Decision	%
Don't wait and start building house	
Wait for the compensation from government in order to build the house	

B. INFORMATION REGARDING CONSTRUCTION MATERIALS

Q9. What kind of wall you are planning to construct in your house?

Type of wall	Frequency
Cement bonded brick wall	
Cement bonded stone wall	
Mud bonded stone wall	
Other1 (*specify*)	

Q10. What kind of roof you are planning to place in your house?

Q11.What all materials required to rebuild your houses is already available to you? *Read name of each material from the list and ask what percentage of total demand can be fulfilled and finally depending upon response tick mark in appropriate box.*

Materials	**Percentage**
Stone	
Brick	
GI sheet	
Wood/plank	
Tile/slate	
Others (Specify)	

Q12. If not available, where will you get all the materials required to rebuild your houses?

%

Type of roof	**Frequency**
GI sheet	
RCC	
Tile/Slate	
Other (*specify*)	

Source of acquiring

Solely purchase from the vendor

Solely use own/natural resources

Both buy and use own/natural resources

C. INFORMATION REGARDING MARKETS

Q13. From which market centers do you prefer to buy non-local and local construction materials? What would be the unit price? How are you planning to transport? How are you planning to procure?

Non-local Materials	Market Name	Distance to travel to reach the market	Approximate Unit Price In NRs	Means of Transportation	Approximate Unit rate of transportation cost in NRs	How do you planning procure? Individually = 1 Collectively = 2

Q14. What are the challenges in purchasing materials in aggregate?

Q 15.What are the other challenges in purchasing construction materials?

Q16. Where do you get the local materials from?

Material	How do you access? (Source)	What means of transportation for buying materials?	Unit rate of transportation cost in Rs
Sand			
Stone			
Wood			
Aggregates			

Q17. What are the difficulties in acquiring local construction materials?

Q.18 If the amount that the government is going to compensate is insufficient then from where will you get the remaining amount to rebuild your house?

Source	**Frequency**
Borrowing from friends	
Selling assets	
Using own deposits	
Remittance	
Bank loan	

Other1 (*specify*)	

D. INFORMATION REGARDING HUMAN RESOURCE AND STORAGE CAPACITY

Q.19 In your opinion, how many skilled manpower are available in this stratum?

Skilled Manpower	**Numbers**
Mason	
Carpenters	
Blacksmiths	

Q20. Where would you store large volume of construction materials?

---END---

Annex-2: Survey questionnaire for transport service providers

Practical Action

Supply Chain of Construction Materials and Tools in Earthquake Affected Districts- Survey 2016/2017

SURVEY WITH TRANSPORTATION SERVICE PROVIDERS

All information asked within this questionnaire will be used only for statistical purposes.

Name of Company/Agency: ______________________

Address: ______________
((District, VDC, Ward)

VAT/PAN No: ______________________

Contact Number: ______________

Owner's Name: ______________________

Place of interview: ______________

Date of Interview: ______________________

Interviewed by: ______________________

Sir/Madam:

Dear respondent,

Practical Action is conducting survey on supply chain of construction materials and tools in earthquake affected districts- *Rasuwa* and *Nuwakot*. In this survey, primarily, situation of infrastructure damage at VDC level, supply and demand situation of different construction materials in the VDC, various issues related to reconstruction will be collected from key informant and will be assessed on the basis of the information obtained. This VDC is selected as a sample for this purpose. The information you provide are confidential by Statistics Act 2015 and they are only published collectively in which no personal records are visible.

I would like to request you to make this survey successful by providing correct and factual information to the enumerators who are coming for survey.

Thanking for your cooperation

Practical Action
South Asia Regional Office
Nepal

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(A) INFORMATION REGARDING GEOGRAPHICAL COVERAGE AND TRANSPORTATION GOODS

Q1. For how long have you been engaged in transportation service sector? (In Years)

Q2. What is your geographical coverage? (Mention name of the place and route)

Route	Road Condition	Route length	Travel time	Main frights		Types of Vehicles and numbers	Fare
				To	From		

Q3. Amongst the goods transported by your vehicle/s, what percent comprises of the construction materials? Please

Q4. Is there any seasonal variation of transportation fare?

Yes ☐ No ☐

Q4. (a). If yes, what is the reason behind the variation? What is the maximum variation?

Q5. Do people insure goods while transporting?

Yes [ ] No [ ]

Q6. While transporting goods, what percent of the good gets damaged compared (%) of the total volume?

Q7. What are the major difficulties in providing regular transportation services?

(B) INFORMATION REGARDING TRANSPORTATION ASSOCIATION

Q8. Is there any transportation association in Rasuwa and Nuwakot district?

Yes [ ] No [ ]

Q9. If yes, please provide details regarding the name of the owner and contact address.

Name of the association	**Contact Person**

Q10. Are you affiliated with the association?

Yes [ ] No [ ]

--END--

Annex- 3: Survey checklist for vendors

Practical Action

Supply Chain of Construction Materials and Tools in Earthquake Affected Districts- Survey 2016/2017

SURVEY WITH VENDORS OF CONSTRUCTION MATERIALS AND TOOLS (BOTH LOCAL AND NON-LOCAL)

All information asked within this questionnaire will be used only for statistical purposes.

Name of enterprise: ____________________ Name of owner ____________

Name of district: ____________________ Market Center: ____________

VAT/PAN number: ____________________ Telephone: ____________

Date: ____________________

Interview Conducted by: ____________________

Sir/Madam:

Dear respondent,

Practical Action is conducting survey on supply chain of construction materials and tools in earthquake affected districts- *Rasuwa* and *Nuwakot*. In this survey, primarily, situation of infrastructure damage at VDC level, supply and demand situation of different construction materials in the VDC, various issues related to reconstruction will be collected from key informant and will be assessed on the basis of the information obtained. This VDC is selected as a sample for this purpose. The information you provide are confidential by Statistics Act 2015 and they are only published collectively in which no personal records are visible.

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LONGITUDE (EAST)			0						

(C) INFORMATION REGARDING VENDOR'S CAPACITY

Q1. How long (in years) have you been selling these materials/tools? Please write the completed years.

Q2. Demand of construction materials is likely to shoot up in future. In this context, can you keep more volume of each material?

Yes No

Q3. If yes, how will you manage to buy additional construction materials?

a) Sufficient personal fund
b) Bank Loan
c) Others: Please specify

Q4. If there is rise in demand for construction materials in future, then does your enterprise have enough capacity to store materials?

Yes No

Q4a. If yes, how and by how much will you increase your enterprise's storage capacity?

Q4b. If no, mention the reason for not being able to increase storage capacity?

Q5. Do you have another construction material shop in Nuwakot or in Rasuwa?

Yes No

Q5a. If you have another such shop, please mention shop's name, address and contact person's name.

Q5b. If no, are you ready to open up a new shop in a market area where demand for construction materials is high?

Yes No

(D) INFORMATION REGARDING CONSTRUCTION MATERIALS

Q6. In past, have you faced any difficulties in purchase or sale of construction materials? Please specify the difficulties

Difficulty	**Activity**	
	Purchase	**Sale**
Producer/Wholesaler		
Transporter		
Payment		
Store		
Labourers		
Others (Specify)		

Q7. What is your mode of payment while purchasing construction materials?

a) Cash

b) Credit

c) Both

Q8. If it's credit, what is your payback period?

Q9. Please mention the name of the bank or the financial institution with whom you are involved with?

Q10. Please mention the construction materials that you sell in the format mentioned below.

CONSTRUCTION MATERIALS		Unit	PARTICULARS												
			Price		Stock		Sales			Source From			Sell To		
			Before EQ	Current	Current	Total capacity	Before EQ	After EQ	Current	Address (District/VDC /Ward)	Mode of Transportation	Cost of Transportation (Per Unit)	Address (District/VDC/Ward)	Mode of Transportation	Cost of Transportation (Per Unit)
Non Local Materials															
Local Materials															
Tools															

Annex- 4: Survey with VDC secretary and key informants

Practical Action

Supply Chain of Construction Materials and Tools in Earthquake Affected Districts- Survey 2016/2017

SURVEY WITH VDC SECRETARY AND KEY INFORMANTS

All information asked within this questionnaire will be used only for statistical purposes.

Name of respondent:: ____________________ Telephone: ______________

Name of district: ____________________ Name of VDC:: ______________

Market Center: ____________________ Stratum: ______________

Date: ____________________

Interview Conducted by: ____________________

Sir/Madam:

Dear respondent,

Practical Action is conducting survey on supply chain of construction materials and tools in earthquake affected districts- *Rasuwa* and *Nuwakot*. In this survey, primarily situation of infrastructure damage at VDC level, supply and demand situation of different construction materials in the VDC, various issues related to reconstruction will be collected from key informant and will be assessed on the basis of the information obtained. This VDC is selected as a sample for this purpose. The information you provide are confidential by Statistics Act 2015 and they are only published collectively in which no personal records are visible.

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(E) INFORMATION REGARDING HOUSE DAMAGES

Q1. How many number of houses used to be in the VDC before the earthquake?

Q2. How many numbers of houses were fully damaged by the earthquake in the VDC?

Q3. How many numbers of houses were partially damaged by the earthquake in the VDC?

Q4. How many numbers of houses were already built after the earthquake in the VDC?

Q5. How many numbers of houses are currently under construction in the VDC?

Q6. How many numbers of partially damaged houses were fixed in the VDC?

Q7 What are the most affected wards?

Settlements (Wards)	Ranking in terms of damage intensity (High, medium ,low)	Economic Status (High, medium low)	Migration (High, medium low)

Q8. Are you aware about the earthquake resistance model houses designed by the government?

Yes [ ] No [ ]

Q9. How many number of households enrolled in the VDC?

Q10. How many households in VDC have signed the agreement?

(F) INFORMATION REGARDING CONSTRUCTION MATERIALS

Q11. What percentage of the houses will likely be made by using following materials in the stratum?

Local materials	% use
Stone	
Brick	
Others (Please Specify …)	

Q12. Please ranked the non-local materials as per demands in future in the stratum?

Note: Please read the names of non-local materials from the list and ask them to rank each material in a four-point scale 1 to 4, where 1 = highest demand and 4 = least demand, and write the ranks against each material.

Non-Local materials	Rank
Cement	
GI sheet	
Iron rebar	
Plane sheet	

Q13. What are the major future infrastructure projects likely to be constructed in the VDCs that may generates demands for construction materials in future?

Note down their responses in brief and to the point in bullet

Q14. Some construction materials such as – sand, stone, brick, timber - are available at the local level. In this context, describe the percentage of demand of the materials that can be fulfilled from the locally available materials.

Note: ***Read name of each local material from the list and ask what percentage of local demand can be fulfilled and finally depending upon response tick mark in appropriate box.***

Local materials	%	How the victims in the VDC access the materials?
Stone		
Brick		
Sand		
Aggregates		
Timber		

Q15. How the victims access the deficit quantity? Describe in detail for each deficit item?

Note down their responses in brief and to the point in bullet

Q16. What are the legal obstacles in acquiring the local materials?

Local materials	**Obstacles (legals and others …)**
Stone	
Brick	
Sand	
Aggregates	
Timber	

Q17. Which is the nearest market center for earthquake victims of this VDC in KM?

Market name:

Distance:

Q18. Are all construction materials available in the nearest market center?

1. All materials are available.
2. Few materials are available.

   Which materials are available?

3. None of the materials are available.

Q19. In your opinion, what could be done to improve the supply system in the nearest market center?

Note down their opinions in brief and to the point in bullet

Q20. What kinds of transportation facilities are available in the stratum for transporting construction materials?

Types of transportation service	% Use	Involvement		
		Male	Female	Child
Light vehicles				
Porters				
Mules				

(G) INFORMATION REGARDING TRAINED MANPOWER

Q21. Please describe the Trained Human resource in the VDC

	Are they available?		Do they meet the demands of the VDC?	
	Yes	No	Yes	No
Mason				
Carpenters				

Blacksmith				
Plumbers				

Q22 In your opinion what could be done to meet the deficit human resources?

Note down their opinions in brief and to the point in bullet

Q23. Please mention the name of agencies who are currently working with their type of works in the VDC?

Name of Agency	Type of Work they are doing	How long has been working?(*write in month*)

---END--

Annex 5: Survey checklist for manufacturer

Practical Action

Supply Chain of Construction Materials and Tools in Earthquake Affected Districts- Survey 2016/2017

SURVEY WITH MANUFACTURERS

All information asked within this questionnaire will be used only for statistical purposes.

Name of Company: ____________________ Contact Person : ________________

Contact Number: ____________________ Interviewed Date:: ________________

Address : ____________________

Interviewed by: ____________________

Sir/Madam:

Dear respondent,

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LONGITUDE (EAST)			0						

A. INFORMATION REGARDING PRODUCTION

Q1. What type of construction materials does your company manufacture? Please name them.

Q2. What is your current production capacity per day?

Q3. What is your maximum production capacity per day?

Q4. Is there any variation in production with regard to various seasons?

Yes No

Q4 (a) If yes, what might be the reasons behind the variation?

Q4 (b) If yes, also specify the production in various seasons.

Q5. What is your storage capacity?

Q6. From where do you source your raw materials?

Q6 (a) Do you have the authority to access the materials?

Yes No

Q6 (b) What difficulties do you face in souring the materials?

Q7 Are there any bottlenecks in operating your business? (*Ask if they have any policy level issue*)

B. INFORMATION REGARDING DISTRIBUTION

Q8. How do you sell your manufactured goods? (*Collect information on Depos, dealers*)

Q9.Provide details on total number, name and location of dealers/depos.

S.No	Name	Location

Q10. If there is sufficient demand then are u willing to open a depo in our project area?

Yes ☐ No ☐

Q10 (a) If yes, what would be the process to open a depo?

Q11. How long will it take to open a depo?

C. INFORMATION REGARDING PRODUCTION

Q12. Do you provide transportation services?

Yes ☐ No ☐

Q12 (a) If yes then provide information on the number and type of vehicle owned?

Type of vehicle owned	Number of vehicles

Q12 (b) If no, what is your mode of transportation and from whom do you avail the transportation services?

Q13. What is the cost of transportation depending upon whether it is owned or has been outsourced by your company?

Location	Own transportation
Narayanghat	
Kathmandu	
Banepa	
Nuwakot	
Rasuwa	

D. INFORMATION REGARDING PRICE

Q14. What is the current price of various materials?

Q15. How is price determined?

Q16. Is price uniform across all districts in Nepal?

Yes ☐ No ☐

Q16 (a) If no then what might be the reason behind price variation?

Q 16 (b) If no then please provide price list for materials across Nepal.

Location	Price

Q17. Has there been change in price after earthquake?

Yes ☐ No ☐

Q17 (a) If yes then by how much and what is the reason behind the hike?

Q18. Do you provide credit facility to buyers?

Yes ☐ No ☐

Q18 (a) If yes, then what is the credit limit?

Q19. Does price decrease with increase in quantity?

E. INFORMATION REGARDING HUMAN RESOURCE

Q20. What is the total number of employees in your organization?

Q21. How many of them are women?

Q22. How many of them are less than 18 years of age?

--END--

Annex 6: List of FGDs Conducted with Location and Participants Detail

Annex7: List of KII with Participants Details

Annex8: The cluster and sub- cluster division

Cluster	Number of VDC/NP by sub-cluster			
	Urban Type	Semi-urban Type	Rural Type	Total
NUWAKOT				
Deurali		3	2	**5**
Ratmate	3		3	**6**
Ranipauwa	2	3		**5**
Kathmandu			4	**4**
Thansing		3	3	**6**
Samundratar		5	4	**9**
Kharanitar		5	1	**6**
Battar	1	13	1	**15**
Betrabrati		4	2	**6**
Total	**6**	**36**	**20**	**62**
RASUWA				
Timure		1	1	**2**
Syafrubesi		1	2	**3**
Thambuchet		2	1	**3**
Dhunche	1		1	**2**
Kalikasthan	1	2	2	**5**
Laharepauwa	1		2	**3**
Total	**3**	**6**	**9**	**18**